

A reproducible benchmark for transformation uncertainty in aquifer-test parameter inference

Ying-Fan Lin^{a,*}

^a*Department of Civil Engineering, Chung Yuan Christian University, Taoyuan, Taiwan*

Abstract

Aquifer-test transmissivity and storage are not direct observations; they are inferred by fitting a response model to drawdown. This paper presents a reproducible computational benchmark for quantifying how that drawdown-to-parameter transformation affects inferred aquifer properties before they enter groundwater models, pumping-capacity calculations, or response-time estimates. Four analytical interpretations—Theis confined flow, Hantush–Jacob leakage, lagging Darcy without leakage, and lagging Darcy with leakage—are fitted to controlled numerical responses and two field pumping tests. A 10,000 scenario groundwater-flow ensemble generated 160,012 fitted interpretation records across homogeneous, leaky, finite-boundary, heterogeneous, and combined settings. The benchmark provides support-matched reference parameters, analytical limiting checks, grid sensitivity checks, and scenario-conditioned model factors. Analytical verification confirmed physical consistency in 22 of 25 limiting cases, and numerical responses aligned with analytical references in 30 of 32 configurations. Benchmark-transfer regressions predicted the 90th percentile absolute log model factor with cross-validated R^2 values of 0.48, 0.51, and 0.55 for transmissivity, storage, and response time. Application to the Massachusetts fractured-bedrock and Lovelock Valley basin-fill cases produced benchmark-conditioned model-factor COV values of 0.61–0.90 and 0.31–0.51, respectively. Model-averaged decision diagnostics showed that the selected interpretation can understate hydraulic capacity and response-time spread. The workflow treats lagging Darcy as one candidate interpretation, not a universal replacement, and reports transformation uncertainty through response residuals, benchmark support, field

*Corresponding author.

Email address: yflin1110@cycu.edu.tw (Ying-Fan Lin)

applicability, and inherited decision consequences.

Keywords: Aquifer testing, Computational hydrogeology, Uncertainty quantification, Numerical benchmark, Model-form error, Reproducible workflow

1. Introduction

Groundwater studies and water resource decisions often depend on aquifer test parameters. Groundwater supply assessment, capture analysis, dewatering design, contaminant migration evaluation, geothermal screening, and regional groundwater models commonly use transmissivity and storage estimates as site evidence (Freeze and Cherry, 1979; Bear, 1979; Anderson et al., 2015). Most site reports derive these parameters by fitting analytical solutions, including classical confined flow, leakage, delayed yield, wellbore storage, and generalized radial flow interpretations, to observed drawdown records (Theis, 1935; Cooper and Jacob, 1946; Hantush and Jacob, 1955; Boulton, 1954; Papadopoulos and Cooper, 1967; Neuman, 1975; Barker, 1988). Advanced hydraulic tomography and geostatistical inversion can then map hydraulic properties across a site or aquifer system (Kitanidis, 1995; Yeh and Liu, 2000; Liu et al., 2007; Illman et al., 2009).

Drawdown is measured directly; transmissivity and storage are inferred through an interpretation model. Stochastic hydrogeology and inverse modeling provide mature tools for propagating parameter uncertainty through flow calculations and site-scale predictions (Dagan, 1989; Gelhar, 1993; Rubin, 2003; Yeh, 1986; Carrera and Neuman, 1986; Beven and Binley, 1992; Hill

20 and Tiedeman, 2007; Doherty, 2015). Groundwater uncertainty research also
21 addresses model structure, conceptual models, hydrostratigraphy, boundary
22 conditions, management forecasts, data worth, and multimodel prediction
23 (Refsgaard et al., 2006; Rojas et al., 2008; Poeter and Anderson, 2005; Ye
24 et al., 2010; Tsai, 2010; Gupta et al., 2012; Tsai and Elshall, 2013; Xu and Val-
25 occhi, 2015; Guillaume et al., 2016; Fienen et al., 2025). Those methods oper-
26 ate after a groundwater model or model ensemble has been defined. Aquifer
27 test interpretation occurs upstream, where model-form error can enter trans-
28 missivity and storage before those values become spatial model inputs, prior
29 distributions, or engineering design parameters. This upstream step matches
30 the broader role of transformation models, which trace derived properties
31 back to the measurements and transformations that produced them (Phoon
32 and Kulhawy, 1999a,b).

33 The upstream interpretation step separates the present work from re-
34 lated uncertainty problems. Slug-test support studies treat test estimates
35 as weighted spatial averages over a near-well support volume (Beckie and
36 Harvey, 2002). Pumping-test upscaling studies ask how interpreted trans-
37 missivity and storativity values should inform regional groundwater model
38 parameters and prior distributions (Manewell et al., 2023, 2024). Aquifer-
39 scale inference studies define when pumping tests can recover effective trans-
40 missivity and storage in heterogeneous aquifers (Naderi and Gupta, 2023).
41 Groundwater BMA and conceptual model studies combine predictions after
42 a model ensemble and parameter estimation strategy has been specified (Li

43 and Tsai, 2009; Tsai, 2010; Tsai and Elshall, 2013). The analysis here targets
44 the earlier conversion from a drawdown record to apparent transmissivity,
45 storage, leakage, and response-time parameters.

46 Groundwater decisions often depend on hydraulic capacity or response
47 time rather than on the fit of a single drawdown curve. Pumping tests are
48 used to size construction dewatering and groundwater control systems. Re-
49 quired pumping capacity scales with transmissivity, and the response time
50 of pressure propagation scales with storage divided by transmissivity (Pow-
51 ers et al., 2007). Groundwater model averaging and management studies
52 show that a single best model can underestimate prediction uncertainty when
53 model structure or parameter estimation methods remain uncertain (Li and
54 Tsai, 2009; Tsai, 2010; Tsai and Elshall, 2013). Aquifer test transforma-
55 tion uncertainty can therefore enter management analysis as under-reported
56 spread in transmissivity, storage, or hydraulic diffusivity.

57 Reported parameter variability can reflect both subsurface heterogeneity
58 and the interpretation process used to estimate parameters. If a classical
59 instantaneous response model omits temporal response structure, such as
60 exchange between fractures and matrix, weakly connected pores, delayed
61 drainage, or lagged storage release (Warren and Root, 1963; Lin and Yeh,
62 2017; Lin et al., 2019), fitting can redistribute the mismatch into transmis-
63 sivity, storage, leakage, or response-time parameters. The resulting param-
64 eter spread can then resemble spatial variability (Sanchez-Vila et al., 1999;
65 Manewell et al., 2023). Groundwater interpretation therefore needs fitted

66 parameters to be reported with log model factor diagnostics, model com-
67 plexity checks, identifiability diagnostics, boundary and leakage alternatives,
68 and validation evidence.

69 The present work develops a reproducible, benchmark-conditioned
70 workflow for aquifer-test transformation uncertainty. Four analytical
71 transformations—Theis confined, Hantush–Jacob leaky, lagging Darcy with-
72 out leakage, and lagging Darcy with leakage—separate classical confined
73 flow, conventional leakage, temporal memory, and combined leakage and
74 memory behavior. Their role is to provide a controlled set of drawdown-to-
75 parameter mappings for estimating how model-form choice changes apparent
76 transmissivity, storage, leakage, and response-time quantities.

77 The evidence chain links response-scale residuals, controlled numerical
78 reference values, reproducible field diagnostics, and groundwater decision
79 quantities. A 10,000 scenario numerical groundwater-flow benchmark pro-
80 vides known reference parameters, analytical limiting behavior, numerical
81 consistency tests, observation-layout variation, and screened model factor
82 distributions. Benchmark-transfer regressions relate the response diagnostics
83 to model-factor magnitude and place the Massachusetts fractured bedrock
84 and Lovelock Valley basin fill field tests on the same conditional uncertainty
85 scale (Cho et al., 2004, 2008; Nadler, 2020a,b). The fitted interpretations are
86 then propagated into model-averaged hydraulic capacity and response-time
87 indices, distinguishing response fitting from the quantities inherited by later
88 groundwater calculations.

89 2. Methods

90 2.1. Transformation uncertainty associated with interpretation models

91 The generic transformation model has the following form (Phoon and
92 Kulhawy, 1999b):

$$\xi_d = \mathcal{T}(\xi_m, \epsilon), \quad (1)$$

93 where ξ_d is the derived or design property, ξ_m is the measured property, $\mathcal{T}(\cdot)$
94 is the generic transformation function, and ϵ is transformation uncertainty.
95 Equation 2 defines the measured property (ξ_m) by integrating a trend (t),
96 inherent variability (w), and measurement error (e):

$$\xi_m = t + w + e. \quad (2)$$

97 A first order second moment approximation expresses the variance of the de-
98 rived property as contributions from inherent variability, measurement error,
99 and transformation uncertainty:

$$\text{Var}(\xi_d) \approx \left(\frac{\partial \mathcal{T}}{\partial w}\right)^2 \text{Var}(w) + \left(\frac{\partial \mathcal{T}}{\partial e}\right)^2 \text{Var}(e) + \left(\frac{\partial \mathcal{T}}{\partial \epsilon}\right)^2 \text{Var}(\epsilon). \quad (3)$$

100 Equations 1, 2, and 3 define the uncertainty vocabulary used here. The
101 two field cases provide response and parameter evidence for an interpreta-
102 tion diagnostic, while complete measurement error budgets and independent

103 heterogeneity measurements remain separate evidence needs. The diagnos-
 104 tic therefore separates field response residuals, apparent parameter spread,
 105 model form support, identifiability, support scale, and decision inheritance.
 106 A full source-by-source uncertainty matrix is provided in Supplementary Ta-
 107 ble S1.

108 2.2. Pumping test interpretation model as a transformation model

109 The measured pumping test vector for observation well i is:

$$\mathbf{y}_{m,i} = \{s_i(t_j), Q, r_i, b, t_j\}_{j=1}^{n_i}, \quad (4)$$

110 where $s_i(t_j)$ is observed drawdown, Q is pumping rate, r_i is radial distance
 111 from the pumping well, b is aquifer thickness, and t_j is elapsed time. The
 112 interpretation model M transforms this measured response into an inferred
 113 parameter vector:

$$\boldsymbol{\theta}_{M,i} = \mathcal{T}_M(\mathbf{y}_{m,i}, \epsilon_{M,i}). \quad (5)$$

114 The classical Darcy interpretation yields:

$$\boldsymbol{\theta}_{D,i} = (S_{D,i}, T_{D,i}), \quad (6)$$

115 whereas the lagging Darcy interpretation yields:

$$\boldsymbol{\theta}_{L0,i} = (S_{L0,i}, T_{L0,i}, \tau_{q,i}, \tau_{s,i}). \quad (7)$$

116 The subscript $L0$ denotes the lagging Darcy interpretation without leakage.

117 The lagging leaky interpretation adds a leakage factor:

$$\boldsymbol{\theta}_{LL,i} = (S_{LL,i}, T_{LL,i}, B_{LL,i}, \tau_{q,i}, \tau_{s,i}), \quad (8)$$

118 with Equation 7 recovered as the diagnostic limit $B_{LL} \rightarrow \infty$. This distinction
119 is important because the original lagging model for a leaky confined aquifer
120 combines aquitard leakage with lagged storage and flow responses (Lin and
121 Yeh, 2017). The present field analysis implements both lagging interpreta-
122 tions and treats B_{LL} , τ_q , and τ_s as fitted response parameters specific to
123 the interpretation model; independent field constraints would be needed to
124 convert them into site specific physical quantities.

125 Equations 4, 5, 6, 7, and 8 make the conversion from drawdown to parameters
126 explicit. The measured quantity is $s_i(t)$, whereas the transformation $\mathcal{T}_M$
127 generates S , T , and any additional response parameters. Unresolved timing
128 structure can remain as a transformation residual if the interpretation model
129 lacks a temporal response component required by the observed response.

130 The fitting process can then compress that residual structure into S and
131 T , producing parameter compensation and inflating interpreted spatial vari-
132 ability across observation wells. The lagging parameters represent temporal
133 memory effects rather than absorbing them entirely into apparent S and
134 T . Here, τ_q is a lumped flux-gradient memory or relaxation time, and τ_s
135 is a lumped storage-release response lag. Both serve as diagnostic response

136 times within the transformation model. The central hydrogeologic question
 137 is how much of the classical drawdown mismatch and interpreted S - T vari-
 138 ability aligns with unresolved temporal dynamics after the analysis includes
 139 classical leakage and boundary alternatives.

140 The transformation model is indexed by interpretation choice. Let a
 141 denote an interpretation model, such as a Theis, leaky, delayed yield, wellbore
 142 storage, generalized radial flow, dual porosity, numerical inverse, or lagging
 143 interpretation. The apparent parameter produced by interpretation model a
 144 is written as

$$\boldsymbol{\theta}_{a,i}^{\text{app}} = \mathcal{H}_a(\boldsymbol{\theta}^*, \mathbf{g}_i, \mathbf{q}, \mathbf{b}, \mathbf{w}_i) + \boldsymbol{\delta}_a(\boldsymbol{\Pi}_i) + \mathbf{e}_{a,i}, \quad (9)$$

145 where $\boldsymbol{\theta}^*$ is the support scale aquifer property that would be needed for an
 146 independent reference test, $\mathbf{g}_i$ is well geometry, $\mathbf{q}$ is the pumping schedule, $\mathbf{b}$
 147 denotes boundary and support assumptions, $\mathbf{w}_i$ is the observation window, $\boldsymbol{\delta}_a$
 148 is the transformation bias specific to interpretation model a and conditioned
 149 on dimensionless controls $\boldsymbol{\Pi}_i$, and $\mathbf{e}_{a,i}$ is residual error associated with the
 150 interpretation model. Because the present field cases lack independent ob-
 151 servations of $\boldsymbol{\theta}^*$, Equation 9 provides diagnostic traceability from measured
 152 drawdown to fitted parameters.

153 *2.3. Field data sets and pumping test operations*

154 *2.3.1. Massachusetts fractured bedrock pumping test*

155 The Massachusetts field case comes from a fractured crystalline bedrock
156 site in eastern Massachusetts, United States (Cho et al., 2004, 2008). The 24
157 hour pumping test occurred in an open bedrock borehole with a 15 cm diam-
158 eter and a 30 m depth, featuring one major water bearing fracture within the
159 upper 23 m. The small bedrock well network samples drawdown responses
160 controlled by fracture connectivity and radial distance from the pumping
161 borehole.

162 The retained Massachusetts subset comprises 12 observation wells and
163 420 drawdown observations. Observation well files record elapsed time and
164 drawdown, while a coordinate table defines the local well layout and distances
165 from the pumping well. Although the original test included variable pumping
166 rate stages (approximately 4, 7, 0, and 2 L/min), this diagnostic treats the
167 retained drawdown records as a constant rate response with $Q = 1.2 \times$
168 $10^{-4} \text{ m}^3 \text{ s}^{-1}$ to match current analysis inputs. This approach standardizes the
169 transformation comparison across wells and defines the Massachusetts case
170 as a diagnostic of the retained constant rate subset. The variable rate stages
171 remain documented as an applicability condition for the retained subset, and
172 the data manifest records the copied raw inputs and their source folder.

173 *2.3.2. Lovelock Valley basin fill pumping/recovery test*

174 The Lovelock Valley field case is multiwell pumping test 2. The analy-
175 sis treats it as an independent basin fill field case separate from the Mas-
176 sachusetts fractured bedrock data. Lovelock Valley is a basin fill valley in
177 the lower Humboldt River Basin of Nevada. The U.S. Geological Survey
178 report describes a valley floor groundwater system bounded and underlain
179 by consolidated rock with low permeability and filled by older and younger
180 alluvial deposits (Nadler, 2020a). The younger basin fill contains the main
181 water bearing deposits: colluvium from the surrounding mountains, fluvial
182 sediments associated with the Humboldt River, and lacustrine clays and silts
183 deposited by Lake Lahontan. The report distinguishes coarser alluvium, flu-
184 vial deposits, and Lahontan clays and silts as hydrogeologic units evaluated
185 with slug tests, multiwell pumping tests, and groundwater modeling (Nadler,
186 2020a). This setting contrasts with the Massachusetts fractured bedrock case
187 because observation wells may respond through layered basin fill aquifers and
188 leaky shallow fine grained units.

189 The associated public data release provides the pumping schedule, well
190 distances, processed drawdown/recovery strings, and PEST observation
191 weights (Nadler, 2020b). Pumping Well 10, an agricultural well completed
192 in the coarser alluvium, discharged at approximately $0.381 \text{ m}^3 \text{ s}^{-1}$ from 28
193 November 2017 10:53 to 4 December 2017 11:28. The retained observation
194 wells span 0.6–9.6 miles from the pumping well and include responses from
195 both coarse alluvial wells and shallow Lahontan clay/silt observation wells.

196 The Lovelock processing follows the structure of the public data release.
197 The response is simulated by superimposing a pumping rate step and a shut-
198 off step so that recovery observations contribute to the fit. The log response
199 offset in Equation 15 is set to 0.05 ft (0.0152 m) to match the report’s thresh-
200 old for drawdown detection. The analysis uses the drawdown/recovery ob-
201 servations, excludes the pumping well, excludes observation strings with zero
202 weight, and retains observation wells whose maximum corrected drawdown
203 exceeds the report’s detection threshold. This leaves seven observation wells
204 and 520 processed observations. Lovelock is analyzed as a separate field case
205 because it differs in site scale, preprocessing, observation weighting, and en-
206 vironmental correction. The retained spatial support for both field cases is
207 reported at the start of the field diagnostic results in Figure 5.

208 *2.4. Analytical interpretation models and fitting protocol*

209 The interpretation protocol applies each field case’s observation records,
210 pumping representation, and well coordinates to four main analytical inter-
211 pretations. Theis confined provides the classical reference without leakage,
212 Hantush–Jacob provides the conventional leaky aquifer reference, lagging
213 Darcy without leakage tests temporal memory without leakage, and lagging
214 Darcy with leakage tests combined leakage and memory behavior. This an-
215 alytical response framing follows aquifer test literature that interprets draw-
216 down through response solutions for confined, leaky, delayed yield, wellbore
217 storage, or generalized radial flow conditions (Theis, 1935; Cooper and Jacob,

1946; Hantush and Jacob, 1955; Boulton, 1954; Papadopoulos and Cooper,
 1967; Neuman, 1975; Barker, 1988). Auxiliary late time regression is used as
 a screening diagnostic outside the field comparison based on the full observa-
 tion record. The lagging responses are evaluated in the Laplace domain, and
 the Stehfest algorithm (Stehfest, 1970) inverts them to the time domain.

For the formal lagging leaky interpretation, the finite radius kernel in the
 Laplace domain is

$$\alpha_{LL}^2(p) = \left(\frac{pS}{T} + \frac{1}{B^2} \right) \frac{1 + p\tau_q}{1 + p\tau_s}, \quad (10)$$

where p is the Laplace variable, α_{LL} is the lagging leaky radial decay term,
 and S , T , B , τ_q , and τ_s are the fitted storage, transmissivity, leakage, flux lag,
 and storage lag parameters for the interpretation model. The corresponding
 constant rate response for a finite radius well is written as

$$\bar{s}_{LL}(p) = \frac{QK_0(\alpha_{LL}r)}{\pi r_w^2 p^2 K_0(\alpha_{LL}r_w) + 2\pi r_w T p \alpha_{LL} K_1(\alpha_{LL}r_w)}. \quad (11)$$

where $\bar{s}_{LL}(p)$ is drawdown in Laplace space, r_w is pumping well radius, r is
 observation distance, and $K_0(\cdot)$ and $K_1(\cdot)$ are modified Bessel functions of the
 second kind. Equations 10 and 11 recover the lagging Darcy interpretation
 without leakage as $B \rightarrow \infty$ and recover the leaky interpretation without lag
 for a finite radius well as $\tau_q = \tau_s = 0$. Numerical checks confirmed these
 limiting cases before field fitting.

For both field cases, nonlinear least squares with multiple starts inde-

236 pendently estimates fitted parameters for each observation well in log pa-
 237 rameter space, applying the log model factor in response (Equation 15) as
 238 the data term. The final refit uses a weak maximum a posteriori (MAP)
 239 objective, appending lognormal prior residuals to the data residuals to avoid
 240 hard boundary solutions in weakly identifiable directions. The prior centers
 241 are $S = 10^{-3}$, $T = 5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, and $\tau_q = \tau_s = 10^3 \text{ s}$. The standard
 242 deviations of the log scale priors are 3.0 and 2.0 for classical S and T , and
 243 2.0, 1.8, 3.0, and 3.0 for lagging S , T , τ_q , and τ_s . Broad fitting bounds re-
 244 main $S \in [10^{-8}, 1]$, $T \in [10^{-8}, 10^{-1}]$, and lag times in $[10^{-4}, 10^9]$ seconds for
 245 the lagging refit without leakage. The lagging leaky field fit uses the same
 246 log response data term, $B \in [1, 10^6] \text{ m}$, and lag time bounds of $[10^{-4}, 10^7]$
 247 seconds, with flags reported when B or a lag direction approaches a bound.
 248 Local MAP tradeoff diagnostics are computed from the numerical Jacobian
 249 of the final log response MAP residual vector at each fitted optimum.

250 The Hantush–Jacob leaky interpretation fits S , T , and leakage factor B
 251 using

$$s_H(t) = \frac{Q}{4\pi T} W\left(u, \frac{r}{B}\right), \quad u = \frac{r^2 S}{4Tt}, \quad (12)$$

252 where $s_H(t)$ is Hantush–Jacob drawdown, $W(u, r/B)$ is the leaky aquifer well
 253 function, and u is the dimensionless time variable. A first type boundary
 254 diagnostic is also retained outside the four main analytical interpretations
 255 to avoid attributing drawdown controlled by boundaries to lagging response

256 alone. This sensitivity check uses an effective recharge image well:

$$s_{CH}(t) = \frac{Q}{4\pi T} \left[E_1 \left(\frac{r^2 S}{4Tt} \right) - E_1 \left(\frac{r_I^2 S}{4Tt} \right) \right], \quad (13)$$

257 where $s_{CH}(t)$ is the constant head image well drawdown diagnostic, $E_1(\cdot)$
 258 is the exponential integral, and r_I is an effective image well distance used
 259 to represent boundary sensitivity. This auxiliary fit tests whether late time
 260 flattening could be explained by an equivalent boundary response outside the
 261 four full record interpretation models.

262 2.5. Transformation model factor in the response

263 At the scale of the measured response, model factors describe transfor-
 264 mation uncertainty when an indirect model converts observations into de-
 265 sign parameters (Phoon and Kulhawy, 1999a,b). Pumping tests require this
 266 idea to retain time and well indices because early, middle, and late time
 267 responses can carry different model errors. Unresolved temporal dynamics—
 268 such as lagged flux or delayed storage release—should appear as structured
 269 drawdown model factors before inverse algorithms interpret the mismatch as
 270 spatial differences in aquifer properties.

271 For observation well i at time t_j , the observed and modeled log responses
 272 are

$$z_{\text{obs},i,j} = \log [s_{\text{obs},i}(t_j) + c], \quad z_{M,i,j} = \log [s_{M,i}(t_j; \boldsymbol{\theta}_{M,i}) + c], \quad (14)$$

273 where $s_{\text{obs},i}$ is observed drawdown, $s_{M,i}$ is drawdown predicted by interpretation
 274 model M , $\boldsymbol{\theta}_{M,i}$ is the fitted parameter vector for well i , and c is a case
 275 specific positive offset that prevents unstable logarithms near zero drawdown.
 276 The Massachusetts analysis uses $c = 0.001$ m, and the Lovelock analysis uses
 277 $c = 0.05$ ft (0.0152 m) following the report's detection threshold. The time
 278 and well indexed log model factor is

$$\eta_{M,i,j} = z_{\text{obs},i,j} - z_{M,i,j} = \ln \left(\frac{s_{\text{obs},i}(t_j) + c}{s_{M,i}(t_j; \boldsymbol{\theta}_{M,i}) + c} \right), \quad (15)$$

279 with the corresponding multiplicative model factor

$$F_{M,i,j} = \exp(\eta_{M,i,j}). \quad (16)$$

280 A value of $F_{M,i,j} = 1$ indicates that the interpretation model reproduces the
 281 observed response at that time and well after fitting. Values above or below
 282 1 indicate observation-model ratios on the drawdown scale. The dispersion
 283 of the log model factor for any diagnostic group G is

$$SD_{\epsilon,M}(G) = \sqrt{\frac{1}{|G| - 1} \sum_{(i,j) \in G} (\eta_{M,i,j} - \bar{\eta}_{M,G})^2}, \quad (17)$$

284 where G may be the full response record, a single well, a field case, or an
 285 early, middle, or late time window; $|G|$ is the number of observations in
 286 that group; and $\bar{\eta}_{M,G}$ is the group mean log model factor. This definition
 287 makes the reported SD_{ϵ} an interpretation dependent and group dependent

288 log model factor dispersion. When a conventional coefficient of variation for
 289 the multiplicative model factor is desired, the lognormal relation is

$$COV_{F,M}(G) = \sqrt{\exp[SD_{\epsilon,M}(G)^2] - 1}. \quad (18)$$

290 The response variance ratio is

$$VR_{\epsilon}(G) = \frac{\text{Var}_{(i,j) \in G}(\eta_{L,i,j})}{\text{Var}_{(i,j) \in G}(\eta_{D,i,j})}. \quad (19)$$

291 Equations 14–19 define the response-level transformation diagnostics.
 292 The analysis reports $SD_{\epsilon,M}(G)$ because the log factor is symmetric for
 293 overprediction and underprediction and remains stable near small drawdown
 294 after adding c . Each well is split into early, middle, and late time windows
 295 using time quantiles within each well to diagnose whether the lagging Darcy
 296 model primarily reduces timing related model factor dispersion.

297 2.6. Variability in inferred hydraulic properties and trend adjustment

298 To diagnose parameter compensation, all positive fitted parameters are
 299 evaluated on the log scale, consistent with common hydrogeologic treatment
 300 of strongly positive and spatially variable hydraulic properties (Dagan, 1989;
 301 Gelhar, 1993; Rubin, 2003). The lognormal coefficient of variation is

$$COV_X = \sqrt{\exp(\sigma_{\ln X}^2) - 1}, \quad (20)$$

302 where X denotes a positive fitted parameter and $\sigma_{\ln X}^2$ is the variance of $\ln X$

303 across wells. Variability in inferred parameters is compared using

$$VR_X = \frac{\text{Var}_i(\ln X_{L,i})}{\text{Var}_i(\ln X_{D,i})}, \quad X \in \{S, T\}. \quad (21)$$

304 Equations 20 and 21 quantify raw variability in inferred hydraulic properties.
305 Interpreting that variability as geologic heterogeneity requires external site
306 evidence in addition to the fitted parameter spread.

307 Raw variance across wells can mix hydrogeologic heterogeneity, observa-
308 tion geometry, and parameter compensation introduced by the model. Vari-
309 ability in inferred hydraulic properties after trend adjustment is evaluated
310 using two simple regressions:

$$\ln X_{M,i} = \beta_{0,M} + \beta_{r,M} \ln r_i + u_{M,i}, \quad (22)$$

311 and

$$\ln X_{M,i} = \beta_{0,M} + \beta_{x,M} x_i + \beta_{y,M} y_i + u_{M,i}. \quad (23)$$

312 where $\beta_{0,M}$ is the intercept, $\beta_{r,M}$ is the coefficient for log distance, $\beta_{x,M}$ and
313 $\beta_{y,M}$ are coordinate coefficients, x_i and y_i are well coordinates, and $u_{M,i}$ is
314 the trend residual. The variance of $u_{M,i}$ is reported as detrended variability
315 in inferred hydraulic properties. Equations 22 and 23 function as diagnostic
316 trend adjustments; a geostatistical representation would require additional
317 spatial support.

318 *2.7. Model complexity, tradeoff, and validation*

319 The candidate interpretation models include different numbers of fitted
 320 parameters, so residual improvement is evaluated alongside model complexity
 321 diagnostics. Approximate AIC and BIC are computed as

$$AIC_M = n \ln \left(\frac{RSS_M}{n} \right) + 2k_M, \quad (24)$$

322 and

$$BIC_M = n \ln \left(\frac{RSS_M}{n} \right) + k_M \ln n, \quad (25)$$

323 where n is the number of response observations, RSS_M is the residual sum
 324 of squares for model M , and k_M is 2 for Theis confined, 3 for Hantush–
 325 Jacob leaky, 4 for lagging Darcy without leakage, and 5 for lagging Darcy
 326 with leakage (Akaike, 1974; Schwarz, 1978). Equations 24 and 25 provide
 327 support for the response model after complexity adjustment, complementing
 328 multiple model and predictive uncertainty practices in groundwater studies
 329 (Poeter and Anderson, 2005; Ye et al., 2010; Doherty, 2015). Local MAP
 330 covariance matrices are used to compute selected correlations among $\ln T_L$,
 331 $\ln S_L$, $\ln \tau_q$, and $\ln \tau_s$, addressing the general calibration need to identify
 332 parameter tradeoffs, compensation, and lack of uniqueness (Yeh, 1986; Hill
 333 and Tiedeman, 2007; Xu and Valocchi, 2015).

334 The validation diagnostic excludes one observation well at a time to test
 335 spatial transfer. The validation diagnostic predicts log parameters for each

336 excluded well from the remaining 11 wells using the log distance trend in
337 Equation 22, and then simulates the excluded well using these predicted
338 parameters. This test follows the predictive logic of cross validation (Vehtari
339 et al., 2017) and focuses on spatial transfer across the retained observation
340 network. It directly tests whether patterns in fitted parameters across wells
341 generalize beyond the specific wells used to define them.

342 *2.8. Numerical groundwater-flow benchmark and analytical verification*

343 Independent values of the true support scale transmissivity, storage, leak-
344 age factor, and response time parameters are unavailable in field pump-
345 ing tests. The main controlled check therefore uses a prescribed numerical
346 groundwater-flow generator before the two field cases are interpreted. MOD-
347 FLOW 6 is used here as a reproducible groundwater-flow solver (Langevin
348 et al., 2017; Bakker et al., 2016), while the method itself applies to any simu-
349 lator that can generate comparable transient drawdown fields. The numerical
350 simulations supply transient responses from Darcy groundwater-flow equa-
351 tions, and the four analytical interpretations then convert those responses
352 into apparent aquifer parameters. This design tests the front-end transfor-
353 mation from drawdown to parameters under controlled numerical aquifer
354 conditions.

355 The numerical ensemble comprises 10,000 fully simulated pumping-test
356 scenarios in ten classes: homogeneous confined flow, distributed vertical
357 leakage, finite no-flow boundary, constant head boundary, heterogeneous

358 hydraulic conductivity, anisotropic heterogeneous hydraulic conductivity,
359 heterogeneous storage, heterogeneous leakage, heterogeneous boundary, and
360 combined leakage and boundary settings (Figure 1). Each class contributes
361 1000 scenarios. The design spans grid spacings of 5.0–10.0 m, square
362 domains of 405–1205 m, six observation layouts with 3–5 wells between
363 near well and far field support, transient records with 20–40 logarithmically
364 spaced output times over 1–10 days, and water level noise levels of 0–0.01
365 m. Heterogeneous cases use correlated lognormal hydraulic conductivity or
366 storage fields, with anisotropy included as a separate class. Each numerical
367 response is then refitted by the four analytical interpretations, generating
368 160,012 fitted interpretation and well records.

369 The reference parameter is defined before fitting. Homogeneous, leaky,
370 and boundary scenarios with uniform properties use the assigned confined
371 transmissivity and storage values. Heterogeneous scenarios use a scenario-
372 specific effective pumping support rather than a fixed radial average. The
373 effective support radius is computed from hydraulic diffusivity, pumping du-
374 ration, maximum observation distance, leakage length where present, and
375 boundary or domain limits. Hydraulic conductivity and storage are then av-
376 eraged with diffusion-decay weights clipped by that effective support, with
377 arithmetic, annular, and fixed-radius sensitivity summaries retained as sec-
378 ondary comparisons. In the 10,000 scenario benchmark, the median effective
379 support radius is 139 m, with a 10th–90th percentile range of 99 m–298 m.
380 This definition is reported explicitly because a numerical cell value is not the

381 same object as a pumping-test parameter.

382 The numerical design defines the benchmark scope through two controlled
383 checks. Closed-form limiting comparisons verify that the implemented ana-
384 lytical models reduce to their expected reference solutions: Hantush–Jacob
385 to Theis as the leakage factor becomes large, lagging Darcy to Theis when
386 response lags collapse, lagging leaky Darcy to lagging Darcy when leakage
387 becomes inactive, and lagging leaky Darcy to Hantush–Jacob when the lag
388 times are equal. Numerical consistency comparisons then relate homoge-
389 neous confined simulations to Theis and distributed leakage simulations to
390 Hantush–Jacob across grid spacings of 2.5–20 m. Model factors are summa-
391 rized for fitted records without active parameter boundary flags and with
392 $\Delta BIC \leq 10$ relative to the best interpretation for the same response. Field
393 use is further screened by applicability criteria that classify each field case
394 as conservative, sensitivity, lower complexity, or non-primary relative to the
395 numerical scenario classes.

396 For each numerical drawdown record, the four analytical interpretations
397 are fitted independently. The response level model factor remains the log
398 factor in Equation 15. The benchmark also defines parameter transformation
399 model factors

$$M_{T,M} = \frac{T_{\text{app},M}}{T_{\text{ref}}}, \quad M_{S,M} = \frac{S_{\text{app},M}}{S_{\text{ref}}}, \quad (26)$$

400 where $T_{\text{app},M}$ and $S_{\text{app},M}$ are apparent parameters obtained by interpretation

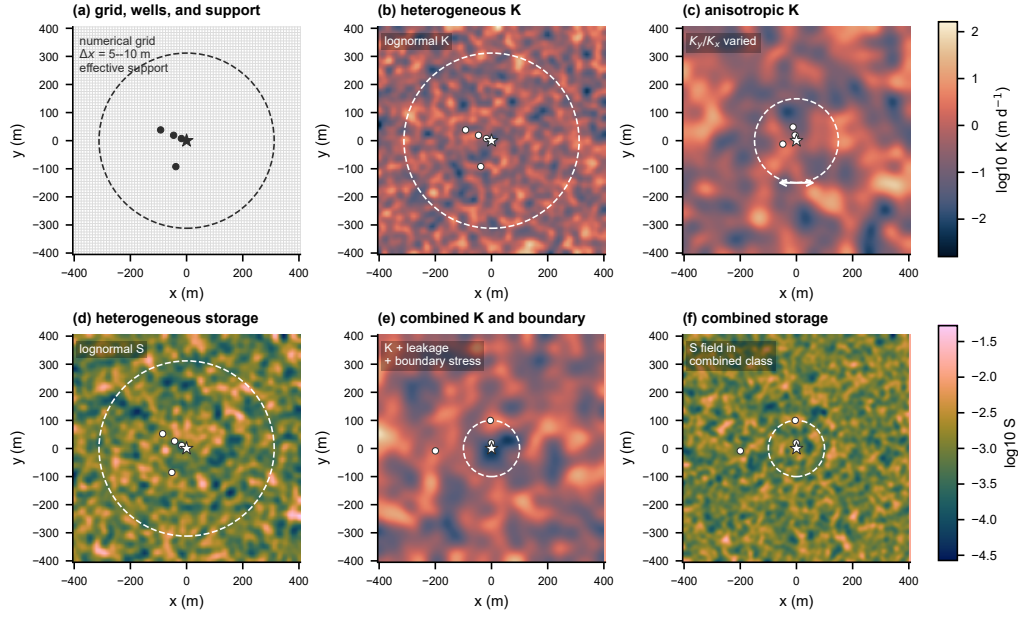


Figure 1: Numerical groundwater-flow benchmark design with MODFLOW 6 through (a) the finite-difference grid, extraction well, observation wells, and scenario-specific effective support envelope for the aquifer-test reference; (b) a representative heterogeneous hydraulic conductivity field; (c) a representative anisotropic hydraulic conductivity field; (d) a representative heterogeneous storage field; (e) combined leakage, boundary, hydraulic conductivity, and storage conditions for transformation stress; and (f) paired numerical response classes for the four analytical interpretations. The numerical fields define controlled drawdown responses for interpretation by the four analytical solutions.

401 model M , and T_{ref} and S_{ref} are the support matched numerical reference
 402 values. The reported model factor coefficient of variation is computed from
 403 the standard deviation of the log model factor:

$$COV_M = [\exp(\sigma_{\ln M}^2) - 1]^{1/2}. \quad (27)$$

404 Equation 27 is used separately for transmissivity, storage, hydraulic control
 405 capacity, and response time. The resulting values are scenario-conditioned
 406 transformation priors applied to field cases when the field diagnostics resem-
 407 ble the benchmark class.

408 The closed-form analytical comparisons are retained as supplementary
 409 controlled checks. They confirm that the implemented analytical solutions
 410 and fitting routines recover clean Theis and leaky responses under prescribed
 411 generating models. The 10,000 scenario numerical benchmark is the main
 412 controlled basis for estimating conditional transformation model factors for
 413 field interpretation.

414 *2.9. Benchmark transfer to field diagnostics*

415 The numerical benchmark provides known reference values, whereas the
 416 field cases provide response and interpretation diagnostics. A benchmark-
 417 transfer analysis estimates the scale of model-factor uncertainty that a field
 418 pumping test may inherit when its diagnostics resemble portions of the bench-
 419 mark. The predictor set is restricted to diagnostics that can be computed
 420 for both numerical and field tests. For benchmark scenario k , the response-

421 diagnostic vector is

$$\mathbf{z}_k = [SD_{\epsilon,M}, p_{\text{BIC},M}, h_M, g_{M,M'}]_{M,M' \in \mathcal{M}}, \quad (28)$$

422 where $SD_{\epsilon,M}$ is the pathway-specific log model factor dispersion, $p_{\text{BIC},M}$ is
 423 the fraction of observation wells for which model M has the lowest BIC,
 424 h_M is the boundary-hit fraction for fitted parameters, and $g_{M,M'}$ is the frac-
 425 tional reduction in SD_{ϵ} between two interpretation models. These quantities
 426 describe the diagnostic behavior of the fitted response.

427 For each numerical scenario, the transferred quantity is the 90th percentile
 428 absolute log model factor for transmissivity, storage, or response time:

$$y_{q,k} = Q_{0.90}(|\ln M_{q,M,i}|), \quad q \in \{T, S, t_c\}. \quad (29)$$

429 A nonlinear regression ensemble $f_q(\mathbf{z}_k)$ is estimated separately for each q
 430 using one row per numerical scenario. Five-fold validation keeps all wells
 431 and interpretation records from a numerical scenario in the same fold. The
 432 fitted relation is then evaluated at the field diagnostic vector $\mathbf{z}_{\text{field}}$ to ob-
 433 tain a benchmark-conditioned estimate $\hat{y}_{q,\text{field}}$. For reporting on a familiar
 434 coefficient-of-variation scale, the 90th percentile absolute log factor is con-
 435 verted to a lognormal-equivalent COV by

$$COV_{q,\text{field}}^{(90)} = \left[\exp \left\{ \left(\frac{\hat{y}_{q,\text{field}}}{1.645} \right)^2 \right\} - 1 \right]^{1/2}. \quad (30)$$

436 Equation 30 defines a benchmark-conditioned sensitivity bracket at the rep-

resented response-diagnostic scale. The regression estimates model-factor magnitude for field diagnostics while the numerical responses and the four analytical interpretations retain separate roles.

2.10. Model-averaged management diagnostic

The management diagnostic tests whether a later groundwater control or management calculation can inherit an overconfident aquifer test input if interpretation uncertainty is collapsed into one fitted parameter set. The calculation follows the Bayesian model averaging logic used in groundwater uncertainty studies, where predictions are combined with posterior model probabilities and uncertainty is separated into components associated with candidate models and parameter estimation methods (Hoeting et al., 1999; Li and Tsai, 2009; Tsai and Elshall, 2013). The model set comprises the four fitted aquifer test interpretations in the present analysis. Their fitted T and S values are propagated into two analytical decision quantities: hydraulic control capacity, which is proportional to T , and characteristic response time, which is proportional to S/T .

For field case or well group i , the posterior weight of interpretation model M is approximated from the BIC values:

$$w_{M,i} = \frac{\exp[-0.5(BIC_{M,i} - BIC_{\min,i})]}{\sum_{M' \in \mathcal{M}} \exp[-0.5(BIC_{M',i} - BIC_{\min,i})]}, \quad (31)$$

where $\mathcal{M}$ contains Theis confined flow, Hantush–Jacob leakage, lagging Darcy without leakage, and lagging Darcy with leakage. Equal prior model

457 probabilities are used so that the weights reflect complexity-adjusted support
 458 from the field response data.

459 Because strict BIC weights can collapse onto one best model, the diag-
 460 nostic also uses a variance window weight. This weighting follows the model
 461 window treatment of Li and Tsai (Li and Tsai, 2009), who showed that fixed
 462 narrow model windows can overrate best models and underestimate multi-
 463 model uncertainty:

$$w_{M,i}^V = \frac{\exp[-0.5\Delta BIC_{M,i}/s_{\Delta BIC,i}]}{\sum_{M' \in \mathcal{M}} \exp[-0.5\Delta BIC_{M',i}/s_{\Delta BIC,i}]}, \quad (32)$$

464 where $\Delta BIC_{M,i} = BIC_{M,i} - BIC_{\min,i}$ and $s_{\Delta BIC,i}$ is the sample standard
 465 deviation of the four ΔBIC values for well or field group i , with a lower
 466 bound of 1. This variance window diagnostic serves as a sensitivity check that
 467 retains plausible analytical interpretations when the information criterion
 468 spread itself is large.

469 For either strict BIC weights or variance window weights, a model-derived
 470 decision variable $x_{M,i}$ is averaged as a discrete BMA distribution:

$$p(x_i | D) = \sum_{M \in \mathcal{M}} w_{M,i}^{(\cdot)} \delta(x_i - x_{M,i}), \quad (33)$$

471 where $w_{M,i}^{(\cdot)}$ denotes either Equation 31 or Equation 32. The two decision
 472 variables are hydraulic diffusivity and characteristic response time:

$$D_{M,i} = \frac{T_{M,i}}{S_{M,i}}, \quad t_{c,M,i} = \frac{r_i^2 S_{M,i}}{4T_{M,i}}, \quad (34)$$

473 where $D_{M,i}$ is hydraulic diffusivity, $t_{c,M,i}$ is the Theis characteristic response
 474 time for observation distance r_i , and $T_{M,i}$ and $S_{M,i}$ are the transmissivity
 475 and storage inferred by interpretation model M . These analytical decision
 476 indices screen the uncertainty inherited by later, project specific groundwater
 477 management calculations.

478 For a hydraulic control or dewatering capacity calculation, the required
 479 capacity scales with transmissivity. If the calculation is based on the selected
 480 interpretation $M_i^* = \arg \max_M w_{M,i}^V$, the robust capacity factor is

$$R_i^C = \max \left[1, \frac{Q_{0.95}\{T_i \mid D\}}{T_{M_i^*,i}} \right], \quad (35)$$

481 and the capacity shortfall probability relative to the selected interpretation
 482 is

$$P_i^C = \Pr(T_i > T_{M_i^*,i} \mid D). \quad (36)$$

483 For monitoring and response planning, the critical risk is that the hydraulic
 484 response arrives earlier than predicted by the selected interpretation. The
 485 response time factor is

$$R_i^t = \max \left[1, \frac{t_{c,M_i^*,i}}{Q_{0.05}\{t_{c,i} \mid D\}} \right], \quad (37)$$

486 with earlier response probability

$$P_i^t = \Pr(t_{c,i} < t_{c,M_i^*,i} \mid D). \quad (38)$$

487 Values greater than one in Equations 35 and 37 indicate that a selected
488 interpretation can pass forward an optimistic capacity or response time es-
489 timate even when other formally fitted interpretations retain support under
490 the variance window BMA.

491 *2.11. Two field cases and scope of inference*

492 The two field cases support a diagnostic validation design with linked re-
493 sponse, transfer, and identifiability checks. Each field case evaluates whether
494 the same four interpretation models reduce structured log model factor dis-
495 persion under its own observation design, pumping representation, and well
496 geometry. The Massachusetts test excludes one observation well at a time
497 to evaluate whether the fitted parameter patterns transfer spatially within
498 the small fractured bedrock network. The Lovelock Valley field case ap-
499 plies the same four interpretation models to an independent basin fill pump-
500 ing/recovery data set and tests field consistency under a different support
501 scale. Boundary flags, leakage support, local MAP correlations, and pa-
502 rameter COV values identify where additional degrees of freedom require
503 independent constraints before direct parameter interpretation.

504 This field design supports diagnostic inference about the interpretation
505 process from drawdown to aquifer parameters. An interpretation model is
506 treated as more strongly supported when log model factor dispersion de-
507 creases, information criteria remain competitive after the parameter penalty,
508 fitted parameters remain away from inactive or boundary values, and valida-

tion evidence is consistent with the fitted response improvement. The practical objective is reporting quality: before S , T , B , τ_q , or τ_s enter groundwater models, dewatering analyses, or management calculations, the interpretation must document how those values were estimated from measured drawdown and how much interpretation dependence remains.

3. Results and Discussion

3.1. Numerical benchmark results

The controlled numerical scenarios establish transformation model factors before those factors are considered for the two field cases. The generating aquifer conditions are prescribed by design, so the numerical ensemble provides the controlled reference basis. Table 1 summarizes the checks, and Figures 2 and 3 show analytical limiting behavior, numerical consistency with analytical responses, and benchmark model factors. The ensemble comprises 10,000 numerical scenarios and 160,012 fitted interpretation and well records; 88,106 records satisfy the screening criteria that exclude active parameter boundary flags and retain interpretations within $\Delta BIC \leq 10$ of the best interpretation for the same response. MODFLOW 6 supplies the response generator, while the four analytical solutions remain the interpretation models.

Verification defines the usable benchmark domain. Analytical limit verification confirmed physical consistency in 22 of 25 cases (Figure 2). The three warnings occur in the finite leakage limiting check, where lagging Darcy with

leakage approaches the Hantush–Jacob response when the two lag times are equal; the largest discrepancy is at 10 m, with a maximum absolute log factor of 0.102 and drawdown root mean square error of 0.0284 m. Numerical consistency checks against analytical reference responses aligned in 30 of 32 cases. The two warnings occur only at 20 m grid spacing for the near observation well in the homogeneous Theis and vertical leakage Hantush generators. The finer 2.5, 5, and 10 m consistency checks aligned with the analytical responses. These results identify near-well finite leakage and coarse-grid settings that require caution when benchmark factors are transferred to field data.

The screened model factor distributions separate response fit from parameter interpretation. Across class and interpretation groups, the median model factor coefficient of variation is 2.06 for transmissivity, 1.27 for storage, and 1.21 for response time, with wider upper tails in leaky, heterogeneous leakage, and combined classes (Figure 3). The empirical distribution functions show how strongly each interpretation can shift the apparent transmissivity, storage, and response time factors away from unity. The distance residual envelope also shows that response residuals remain observation-support dependent. Variance attribution within the screened records confirms the separation: response residual standard deviation is dominated by water-level noise, followed by interpretation model and observation distance, whereas transmissivity and storage model factors are led by interpretation model and only then by scenario class, observation distance, or noise level.

Table 1: Numerical groundwater-flow benchmark for field-inference scope via MODFLOW 6 as the controlled response generator, scenario-conditioned transformation model factors for the four analytical interpretations, and field conditions under which benchmark factors can inform field diagnostics.

Component	Numerical benchmark result	Field interpretation
Coverage	10,000 numerical scenarios in ten classes; six observation layouts; 160,012 fitted interpretation and well records; 88,106 screened records.	Provides controlled response generation before field evidence is interpreted.
Verification	Analytical limit verification confirmed physical consistency in 22 of 25 cases; numerical consistency checks aligned with analytical reference responses in 30 of 32 cases; remaining warnings identify near well leakage and coarse grid limits.	Numerical support and discretization effects remain visible and should be reported.
Model factors	Screened class and interpretation median COV values are 2.06 for transmissivity, 1.27 for storage, and 1.21 for response time; upper tails remain scenario dependent.	Conditional model factor COV values apply to matching benchmark classes.
Applicability	Massachusetts receives a primary conservative match for combined leakage and boundary behavior; Lovelock receives primary matches for heterogeneous storage and hydraulic conductivity.	Field geometry and true support scale define the applicable range of the scenario-conditioned values.

554 These numerical results are used as scenario-conditioned transformation
555 factors. Field application therefore occurs only after response, residual, com-
556 plexity, and parameter-spread diagnostics identify a comparable benchmark
557 class. This sequencing keeps the benchmark factors conditional on observed
558 field evidence rather than treating them as generic groundwater design COV
559 values.

560 *3.2. Benchmark-conditioned transfer to field diagnostics*

561 The benchmark-transfer regression connects the controlled benchmark to
562 field use while preserving ambiguity among possible aquifer-process families.
563 Cross validation by complete numerical scenario shows that response diag-

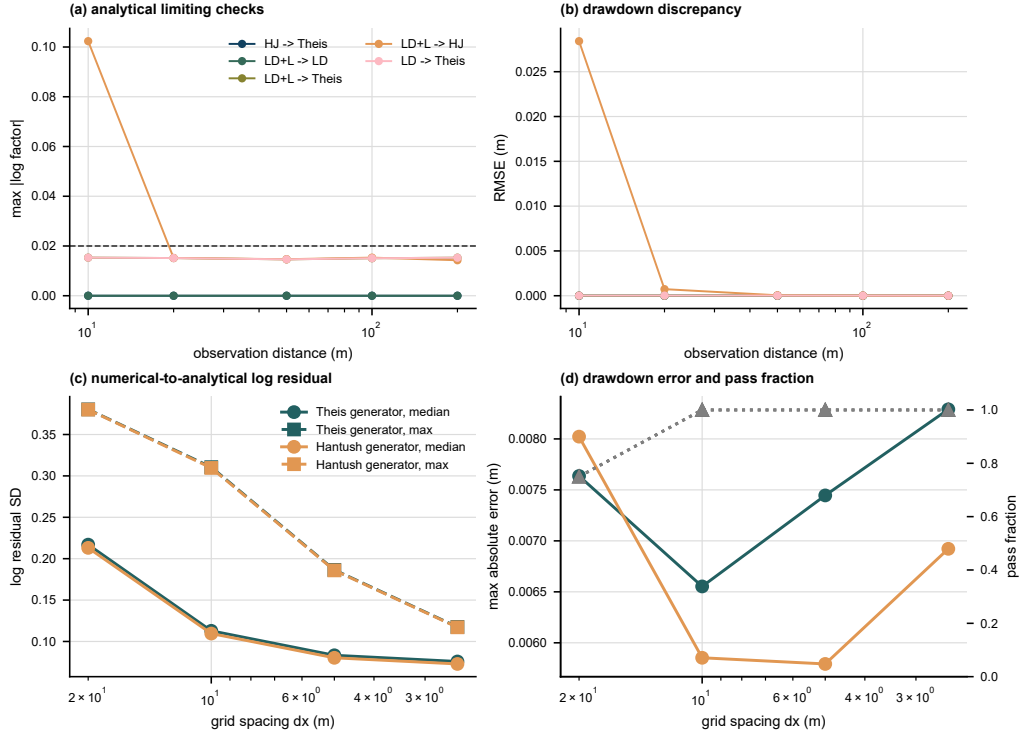


Figure 2: Analytical verification and numerical consistency checks through (a) analytical limit behavior versus observation distance and the dashed log factor tolerance for limit checks; (b) additional analytical limit behavior versus observation distance; (c) numerical consistency with homogeneous Theis analytical reference responses across grid intervals; and (d) numerical consistency with vertical leakage Hantush analytical reference responses across grid intervals. MODFLOW 6 numerical responses and caution labels define finite leakage, near-well, and coarse-grid limits for benchmark scope.

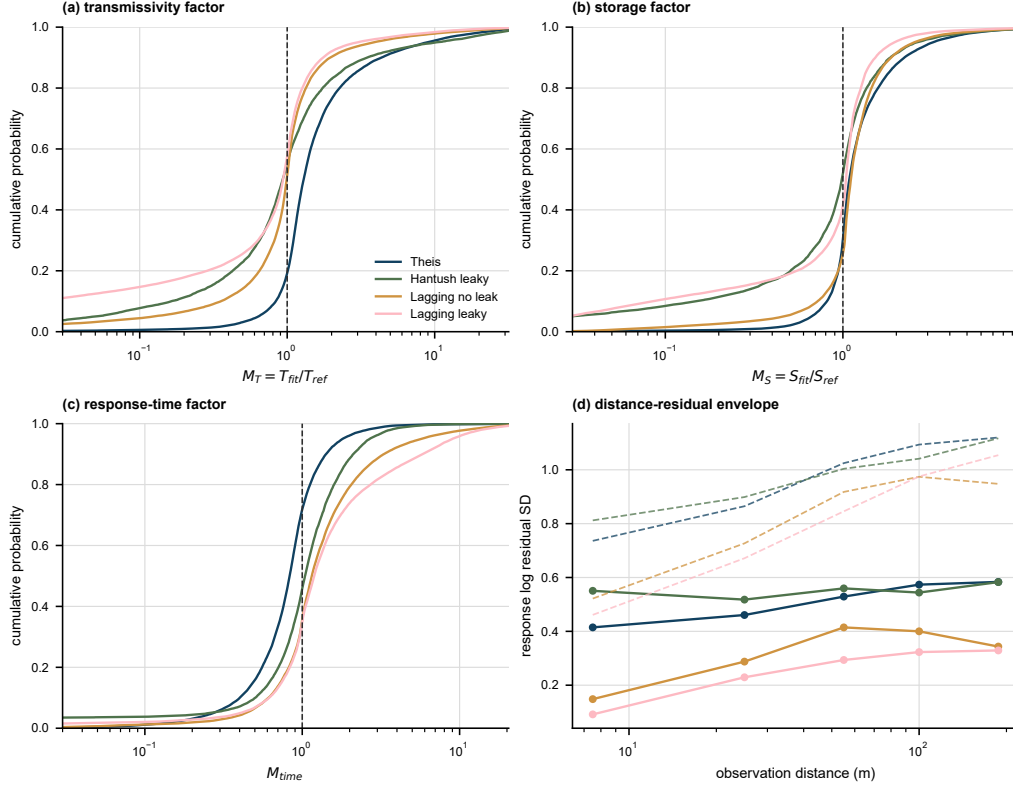


Figure 3: Screened numerical benchmark model factors through (a) empirical distribution functions for transmissivity model factors after boundary-active fit removal and retention of interpretations within $\Delta BIC \leq 10$; (b) empirical distribution functions for storage model factors under the same screened records; (c) empirical distribution functions for response-time model factors under the same screened records; and (d) median response log residual standard deviation by observation distance, with dashed lines for the 90th percentile envelope. The vertical dashed line in (a)–(c) denotes a factor of one, and all values represent scenario-conditioned model factors for the benchmark classes.

564 nostics predict the 90th percentile absolute log model factor with $R^2 = 0.48$
 565 for transmissivity, $R^2 = 0.51$ for storage, and $R^2 = 0.55$ for response time
 566 (Figure 4). The strongest predictors are lagging-response residual dispersion,
 567 boundary-hit fractions, and BIC preference fractions, indicating that trans-
 568 fer is driven by the same diagnostics used to evaluate the four analytical
 569 interpretations.

570 The field transfer is therefore reported as a model-factor scale across com-
 571 parable benchmark diagnostics. For Massachusetts, the benchmark transfer
 572 gives p90-equivalent model-factor COV values of 0.61 for transmissivity, 0.65
 573 for storage, and 0.90 for response time. For Lovelock Valley, the correspond-
 574 ing values are 0.51, 0.31, and 0.36. These values are smaller than several
 575 class-specific conservative COV values in the deterministic applicability ta-
 576 ble because the regression averages across similar diagnostic behavior in the
 577 full numerical ensemble. They are used as benchmark-conditioned sensitivity
 578 brackets alongside the deterministic applicability criteria.

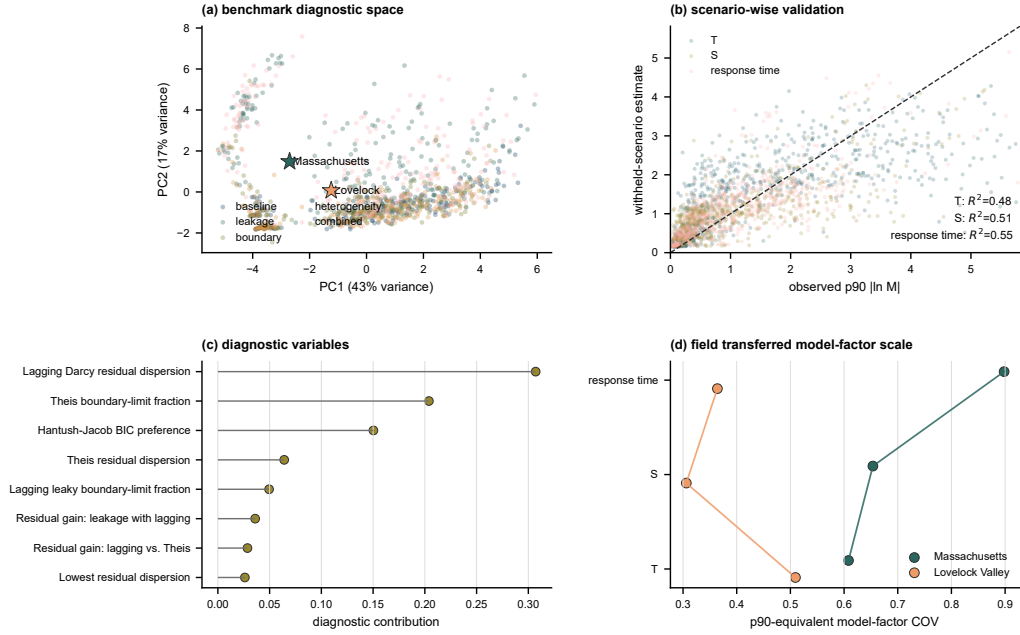


Figure 4: Benchmark transfer of aquifer-test transformation uncertainty to field diagnostics through (a) the numerical benchmark response-diagnostic space and the positions of the two field cases; (b) cross-validated predictions versus observed 90th percentile absolute log model factors for transmissivity, storage, and response time; (c) the main diagnostic variables for the regression; and (d) benchmark-transferred, 90th-percentile-equivalent model-factor COV values for the two field cases.

579 *3.3. Massachusetts diagnostic results*

580 The field evidence begins with the observation support of the two real
581 cases (Figure 5). Massachusetts observation wells span 5.8–36.6 m from the
582 pumping well and provide 420 retained drawdown observations. Drawdown
583 reproduction provides the primary Massachusetts response evidence: the four
584 interpretation models are overlaid for all 12 observation wells in Figure 6.

585 Lagging response terms sharply reduce the Massachusetts log model
586 factor dispersion. Across the four interpretations based on the full ob-
587 servation record, pooled SD_ϵ decreases from 0.166 for Theis confined to
588 0.164 for Hantush–Jacob leaky, 0.062 for lagging Darcy without leakage,
589 and 0.035 for lagging Darcy with leakage. The largest reduction occurs
590 after the model includes a lagging response term, whereas the leakage term
591 still requires support from complexity and parameter-variability evidence.
592 Observed-predicted scatter and pooled log model factor distribution plots
593 are retained as Supplementary Figures S1 and S2.

594 Model complexity and parameter variability qualify this response result
595 (Figure 7). BIC prefers lagging Darcy with leakage in 5 of 12 wells, lagging
596 Darcy without leakage in 3 wells, Hantush–Jacob leaky in 2 wells, and Theis
597 confined in 2 wells. The transmissivity COV values are 0.58, 0.78, 0.46, and
598 0.81, and the storage COV values are 2.05, 1.71, 1.74, and 1.54 for the same
599 model order. Thus, the Massachusetts case supports lagging response as a
600 transformation diagnostic, while the leaky lagging model requires identifi-
601 bility checks before its additional parameter is interpreted as a field property.

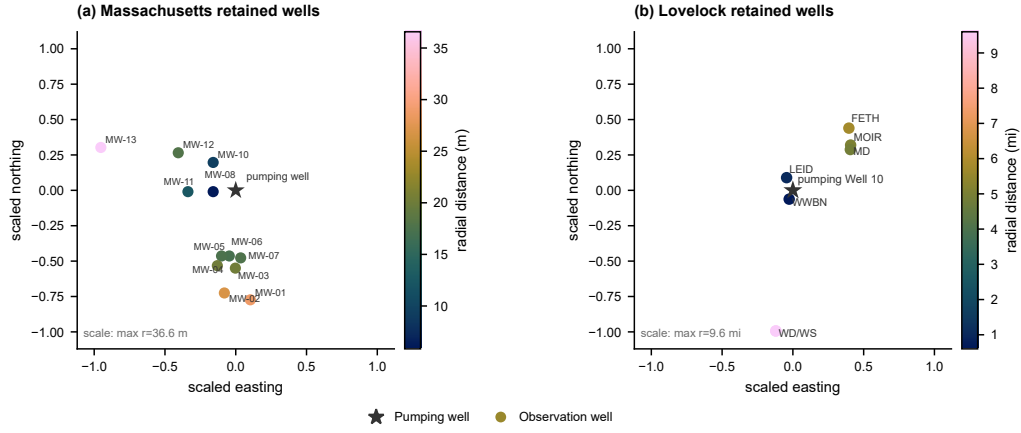


Figure 5: Paired field-data spatial support for the two real cases through (a) Massachusetts fractured bedrock retained observation wells; and (b) Lovelock Valley MWP2 retained observation wells relative to extraction Well 10. Both panels use coordinates centered on the extraction well and normalized by the maximum retained radial support within the case, so panel shape denotes observation geometry rather than absolute field scale. Circles denote retained observation wells, stars denote extraction wells, and colors retain the actual radial distance within each case.

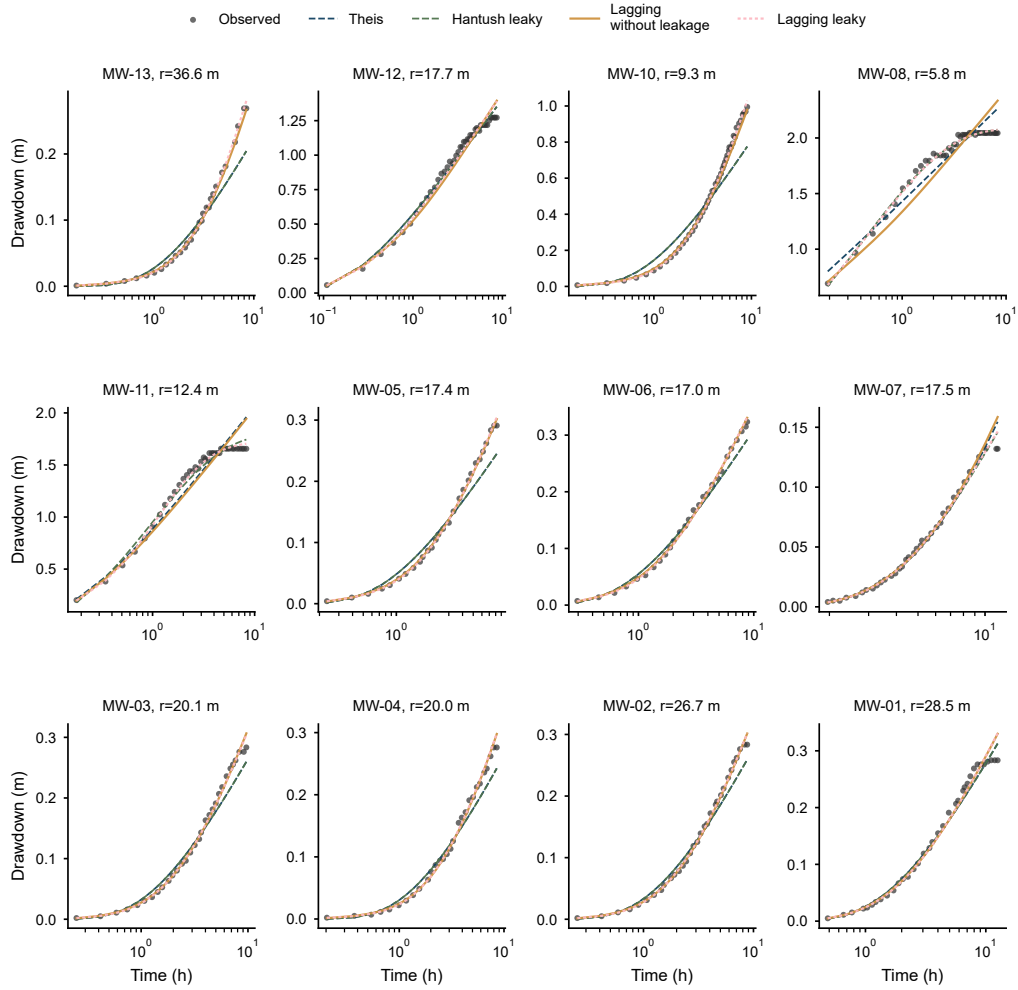


Figure 6: Massachusetts observed and simulated drawdown curves for the 12 observation wells across the four main interpretation models, with measured drawdown as points and Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, and lagging Darcy with leakage predictions as lines.

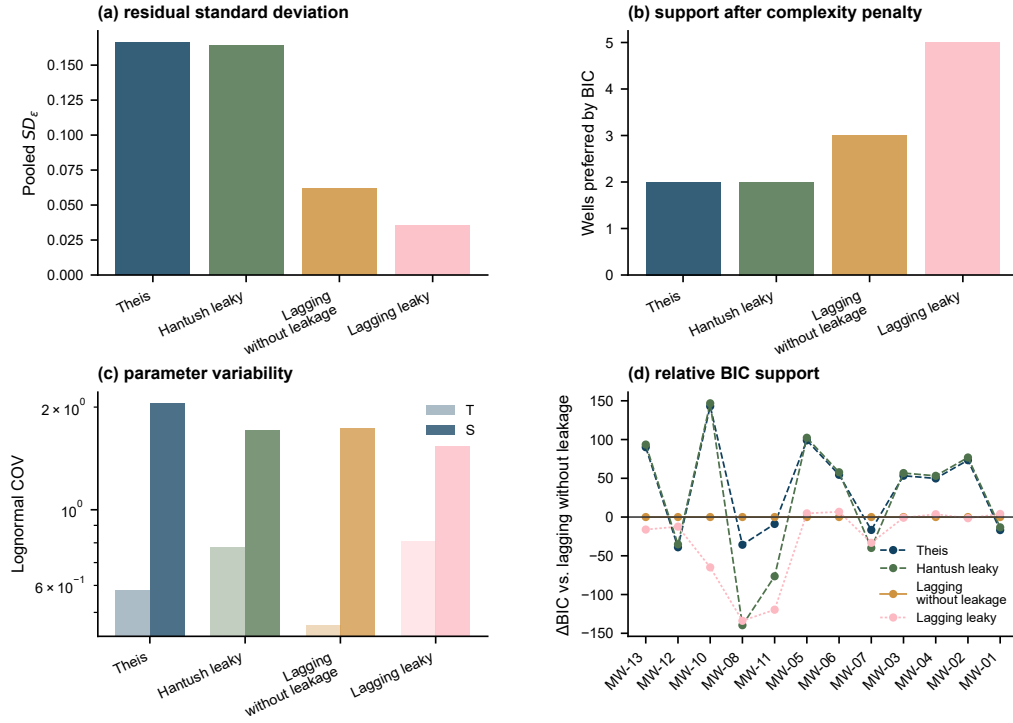


Figure 7: Massachusetts interpretation-model summary through (a) pooled log model factor dispersion; (b) BIC preference counts across the four main interpretation models; (c) lognormal parameter COV for fitted transmissivity and storage; and (d) relative BIC support against lagging Darcy without leakage.

602 3.4. Lovelock Valley diagnostic results

603 Lovelock Valley provides the independent basin fill field case. The
604 retained records include seven observation wells, 520 processed pump-
605 ing/recovery observations, a larger field support of 0.6–9.6 miles, public
606 observation weights, and a 0.05 ft offset in the log response tied to the
607 report’s detection threshold. The Lovelock response curves mirror the
608 Massachusetts drawdown comparison and show fitting at the response level
609 before fitted parameters are interpreted (Figure 8).

610 Lovelock applies the same response-level evidence chain used for Mas-
611 sachusetts. Across the seven selected observation wells, pooled SD_ϵ is 0.246
612 for Theis confined, 0.294 for Hantush–Jacob leaky, 0.174 for lagging Darcy
613 without leakage, and 0.174 for lagging Darcy with leakage. The log model
614 factor evidence again favors a lagging interpretation, but the leaky model
615 remains indistinguishable from the lagging model without leakage at the
616 response level. The matching observed-predicted scatter and model factor
617 distribution plots are provided as Supplementary Figures S3 and S4.

618 Complexity-adjusted Lovelock support is concentrated in interpretations
619 without leakage (Figure 9). BIC prefers lagging Darcy without leakage in
620 5 of 7 wells and Theis confined in 2 of 7 wells. Transmissivity COV val-
621 ues are 3.40, 3.70, 1.83, and 4.77, and storage COV values are 0.89, 0.86,
622 0.95, and 10.21 in the same model order. Thus, Lovelock supports lagging
623 response structure but provides no complexity-adjusted support for an addi-
624 tional leakage parameter.

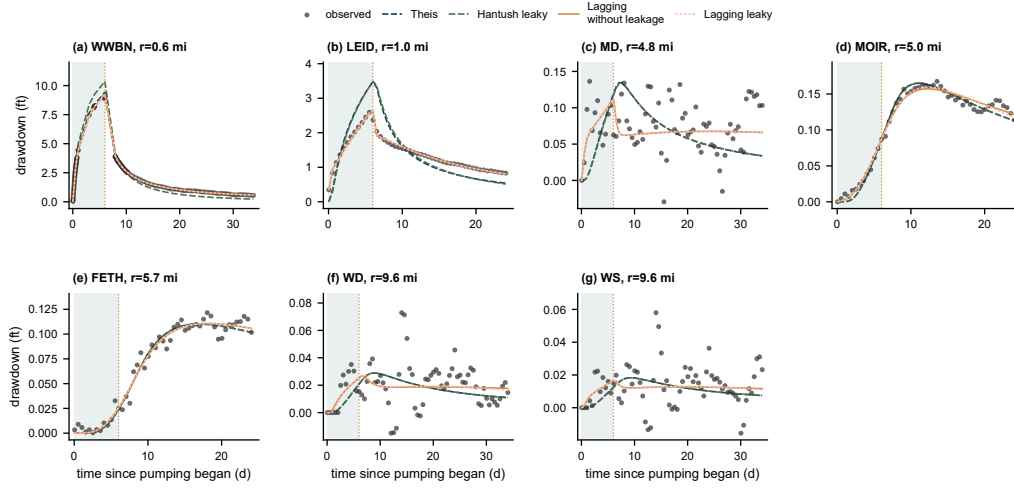


Figure 8: Lovelock Valley MWP2 observed drawdown and recovery responses with fitted curves under the four main interpretation models across seven selected observation wells by distance from Well 10, with measured responses as points, four model predictions as lines, the shaded interval as the extraction period, and the dotted line as shutoff.

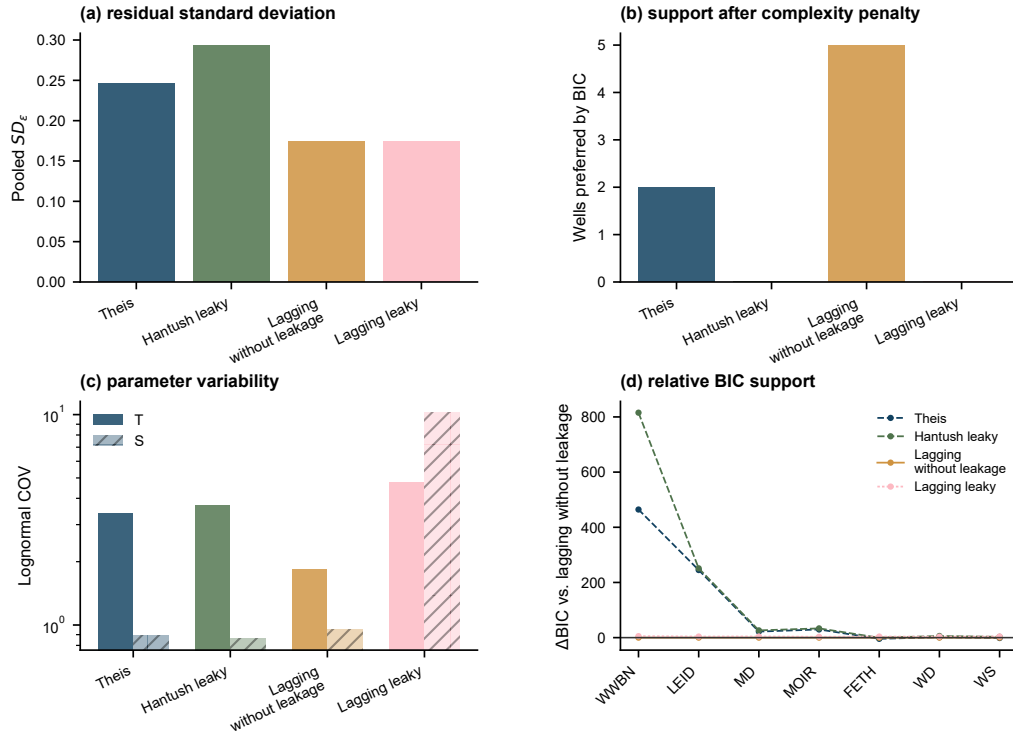


Figure 9: Lovelock Valley interpretation-model summary through (a) pooled log model factor dispersion; (b) BIC preference counts across the four main interpretation models; (c) lognormal parameter COV for fitted transmissivity and storage; and (d) relative BIC support against lagging Darcy without leakage.

625 *3.5. Synthesis of the two field cases and scope of inference*

626 The two field cases show parallel response evidence but different leak-
627 age support. Massachusetts represents a small fractured bedrock drawdown
628 subset with 12 wells, 420 retained observations, and 5.8–36.6 m radial sup-
629 port. Lovelock Valley represents a larger basin fill pumping/recovery case
630 with seven retained observation wells, 520 processed observations, 0.6–9.6 mi
631 radial support, and public weights tied to the report’s detection threshold.
632 In the common Theis / Hantush–Jacob / lagging Darcy without leakage /
633 lagging Darcy with leakage order, Massachusetts gives $SD_\epsilon = 0.166 / 0.164$
634 $/ 0.062 / 0.035$, BIC preferred wells = 2 / 2 / 3 / 5, and transmissivity COV
635 = 0.58 / 0.78 / 0.46 / 0.81; Lovelock Valley gives $SD_\epsilon = 0.246 / 0.294 /$
636 $0.174 / 0.174$, BIC preferred wells = 2 / 0 / 5 / 0, and transmissivity COV
637 = 3.40 / 3.70 / 1.83 / 4.77. The full four-model headline table, including
638 storage COV and validation scope, is provided in Supplementary Table S2.

639 The final synthesis identifies transformation uncertainty as a reporting is-
640 sue in aquifer test interpretation (Figure 10). In both field cases, the lagging
641 interpretation reduces log model factor dispersion relative to the classical
642 confined reference. Direct geological interpretation of additional fitted pa-
643 rameters requires the accompanying diagnostics for response support, com-
644 plexity support, parameter spread, and validation scope. Fitted transmissiv-
645 ity and storage should therefore be reported with the interpretation model,
646 log model factor dispersion, complexity support, parameter spread, and the
647 scope of validation for each case.

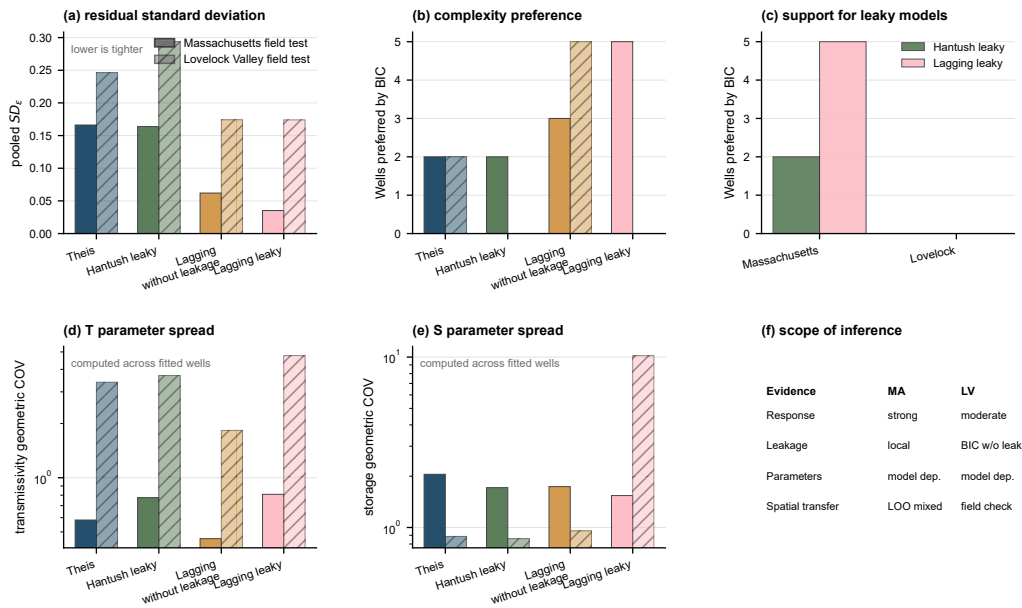


Figure 10: Validation synthesis from the Massachusetts fractured bedrock case and the Lovelock Valley basin-fill case through (a) pooled log model factor dispersion; (b) BIC preference counts by well; (c) support for leakage-inclusive models; (d) transmissivity COV; (e) storage COV; and (f) inference scope across response support, leakage support, parameter interpretation, and spatial transfer availability.

648 After the two field evidence chains are established, field applicability
649 screening links the field diagnostics back to the numerical benchmark classes
650 (Figure 11). Massachusetts maps primarily to the combined leakage and
651 boundary class, with heterogeneous leakage, heterogeneous boundary, and
652 heterogeneous hydraulic conductivity classes retained as sensitivity or lower
653 complexity checks. Lovelock Valley maps primarily to heterogeneous stor-
654 age and heterogeneous hydraulic conductivity classes, while heterogeneous
655 leakage is retained only as a local leakage sensitivity because BIC support
656 concentrates in no leakage interpretations. These criteria define where bench-
657 mark model factors can inform field interpretation for each site.

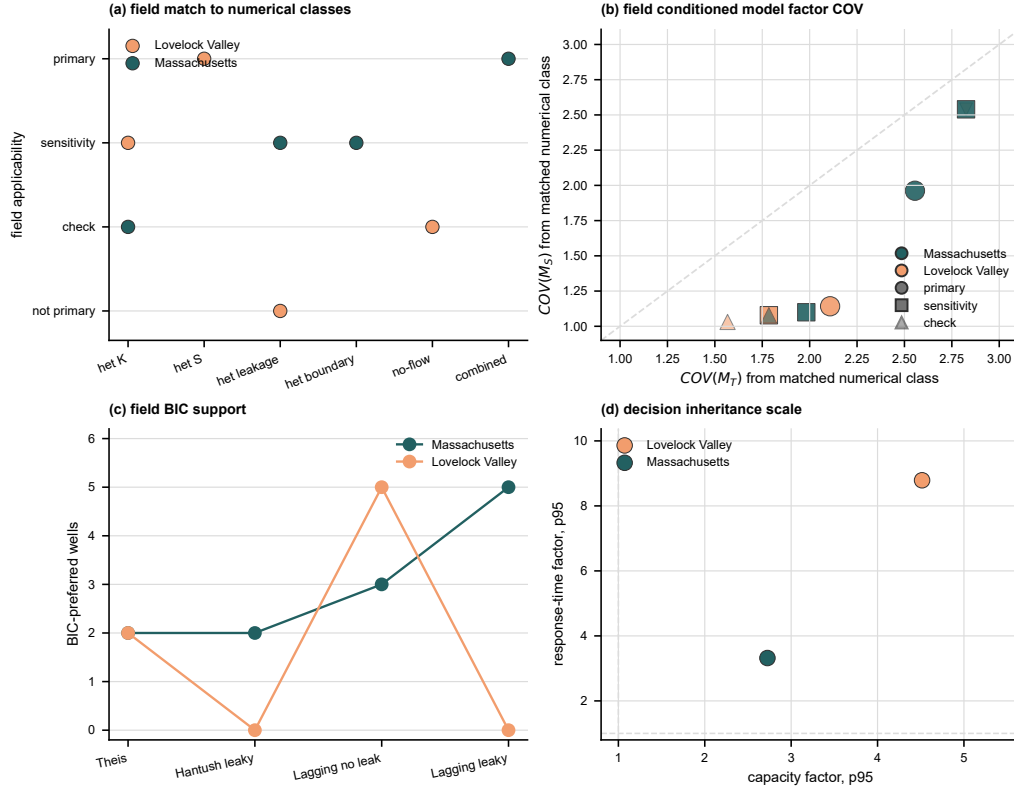


Figure 11: Field application of scenario-conditioned transformation model factors after field evidence-chain establishment through (a) assignment of the two field cases to numerical benchmark classes and applicability levels; (b) matched transmissivity and storage model factor COV values as field sensitivity brackets or conservative priors; (c) field BIC support for the four analytical interpretations; and (d) inherited 95th percentile capacity and response-time factors from the model-averaged decision diagnostic. The figure identifies where benchmark factors enter the field interpretation.

658 *3.6. Management consequence of aquifer test transformation uncertainty*

659 The model averaged management diagnostic carries the four aquifer test
660 interpretations into a BMA distribution upstream of a site-scale groundwater
661 model. Figure 12 reports decision quantities derived from the same fitted T
662 and S values, and Supplementary Table S3 gives the numerical table. Strict
663 BIC weights represent single-model reporting when the information criterion
664 concentrates support on one interpretation. Variance window weights retain
665 plausible alternatives by normalizing BIC differences by the well specific BIC
666 spread.

667 The result shows why the engineering consequence is better expressed
668 through transmissivity, hydraulic diffusivity, and response time. Under strict
669 BIC weights, the median robust hydraulic control capacity factor is 1.00 in
670 both field cases, reflecting the near single model behavior of the posterior
671 weights. Under variance window weights, the median capacity factor in-
672 creases to 1.60 in Massachusetts and 1.18 in Lovelock Valley, with 95th per-
673 centile factors of 2.73 and 4.52. The corresponding mean probability that
674 the selected interpretation underestimates hydraulic control capacity is 0.38
675 in Massachusetts and 0.21 in Lovelock Valley.

676 The response time consequence is also larger than the observation-scale
677 drawdown difference. Under variance window weights, the median response
678 time factor is 1.85 in Massachusetts and 4.05 in Lovelock Valley, with 95th
679 percentile factors of 3.32 and 8.79. The mean probability that the BMA
680 mixture contains a faster response than the selected interpretation is 0.41

681 and 0.23. These values are aquifer-test inheritance diagnostics rather than
682 project failure probabilities because they are computed from interpretation
683 models rather than a project specific MODFLOW management model. They
684 show what a later groundwater control or management calculation can inherit
685 if a pumping test report provides one selected T and S without documenting
686 interpretation model uncertainty.

687 The model weights also show that the decision diagnostic is not driven
688 by always selecting the most complex interpretation. Massachusetts gives
689 nonzero variance window support to all four interpretations, with the largest
690 mean weight on lagging Darcy with leakage. Lovelock Valley assigns the
691 largest mean weight to lagging Darcy without leakage, but Theis and lag-
692 ging with leakage both retain nontrivial support. The diagnostic therefore
693 uses the field supported mixture in each case rather than forcing the same
694 preferred interpretation onto both sites. The key inheritance problem is that
695 strict single-model reporting can make the aquifer test input look sufficiently
696 certain even when alternative formal interpretations would change capacity
697 or response time decisions by factors of two to nine.

698 *3.7. Implications for groundwater uncertainty analysis*

699 Aquifer test parameters used in groundwater models and management
700 calculations depend on the interpretation model that converts drawdown to
701 fitted values. That transformation changes both the log model factor in the
702 response and the inferred parameter field. The evidence therefore combines

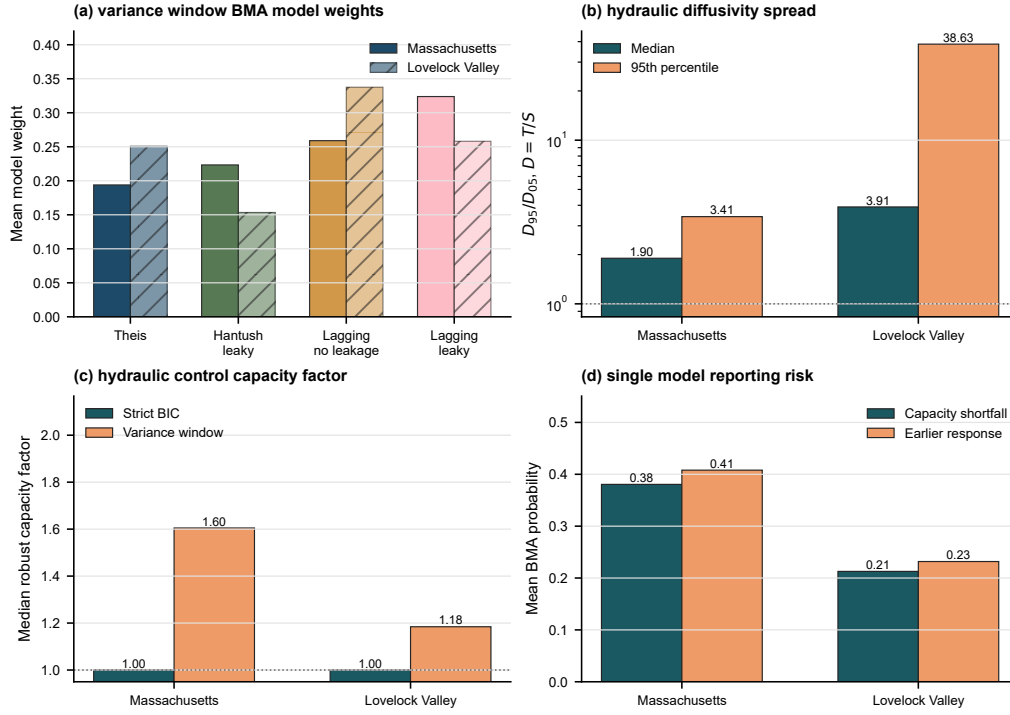


Figure 12: Model-averaged propagation of aquifer-test interpretation uncertainty to management-relevant hydraulic quantities through (a) mean variance-window BMA weights for the four analytical interpretations; (b) median and 95th percentile hydraulic diffusivity spread, expressed as D_{95}/D_{05} with $D = T/S$; (c) median robust hydraulic control capacity factor under strict BIC weights and variance-window weights; and (d) mean variance-window BMA probability of selected-interpretation underestimation of hydraulic control capacity or missed earlier hydraulic response.

703 controlled numerical checks under prescribed aquifer conditions, log model
704 factor dispersion, information criteria, parameter variability, leakage support,
705 field case limitations, and two-case consistency. The numerical benchmark
706 tests recovery and model factor calibration where the aquifer conditions are
707 controlled by design. Site specific field diagnostics support local interpreta-
708 tion, while the cross-case inference rests on evidence that appears in both
709 field cases. The resulting reporting endpoint is to state the measured re-
710 sponse support, interpretation model, response log model factor, complexity
711 support, parameter spread, numerical benchmark match, field applicability
712 screening, and decision consequence before pumping-test-derived hydraulic
713 properties are used; the full checklist is given in Supplementary Table S4.

714 The reporting structure addresses a central groundwater interpretation
715 risk. If model-form error is hidden inside fitted transmissivity and storage
716 values, later groundwater flow models, capture analyses, dewatering calcu-
717 lations, or allocation studies may treat transformation error as site hetero-
718 geneity. Reporting the log model factor, the complexity-adjusted support,
719 the inferred parameter spread, and the validation scope makes that distinc-
720 tion explicit. These diagnostics define a stricter standard for deciding when
721 aquifer test parameters are ready to support groundwater interpretation.

722 *3.8. Lagging Darcy as a diagnostic tool*

723 The lagging Darcy interpretations reduce log model factor dispersion.
724 This reduction directly addresses the transformation uncertainty term in

Equation 5 because SD_ϵ is the group dispersion of $\eta_{M,i,j}$ defined in Equation 17. The classical transformation without lag leaves a systematic response mismatch, making the log model factor structure a signal from the interpretation model before inverse algorithms treat it as aquifer heterogeneity. Lagging Darcy therefore serves as a diagnostic interpretation for nonsteady response structure within the broader set of established aquifer test interpretations. Its no-leakage and leaky forms separate two questions: whether temporal memory is needed, and whether leakage is supported strongly enough to justify a finite B . Groundwater flow models, risk assessments, and engineering calculations frequently inherit T and S values from aquifer test interpretations; if response mismatches introduced by the model remain hidden inside those inferred parameters, later spatial analyses may interpret transformation error as hydrogeologic heterogeneity.

3.9. Leakage and first type boundary alternatives

The diagnostic for late time flattening remains important because it tests conventional explanations alongside lagging behavior. Hantush–Jacob leakage and a first type constant head boundary can both flatten drawdown at late time, while Theis retains the familiar confined aquifer reference interpretation. In the Massachusetts case, auxiliary checks indicate selective leakage or behavior equivalent to a boundary, with support concentrated in specific wells. In the Lovelock case, BIC preference is concentrated in interpretations without leakage. The result across cases supports treating leakage as an ex-

747 plicit competing interpretation before fitted storage and transmissivity are
748 interpreted as direct evidence of aquifer heterogeneity.

749 The methodological implication is that leakage and boundary alternatives
750 should remain explicit candidates. The original lagging leaky model is rep-
751 resented in the field comparison, and the leakage factor carries interpretive
752 value when it is identifiable and improves the field evidence. The compari-
753 son therefore treats lagging with leakage as a more demanding interpretation
754 than the lagging interpretation without leakage.

755 *3.10. Why response improvement and parameter convergence differ*

756 The inferred parameters place the response improvement in context. Lag-
757 ging Darcy reduces log model factor dispersion, but parameter convergence is
758 weaker and differs by site. This pattern aligns with aquifer test studies show-
759 ing that drawdown-inferred parameters often behave as apparent or effective
760 quantities, with transmissivity and storage responding differently to hetero-
761 geneity, observation geometry, and model form (Sanchez-Vila et al., 1999;
762 Manewell et al., 2023). The central diagnostic result is this contrast between
763 response fit and parameter interpretation. A model can reproduce draw-
764 down more accurately while redistributing uncertainty among S , T , B , τ_q ,
765 and τ_s . Lagging Darcy reports should include log model factor dispersion,
766 information criteria, parameter COV, leakage support, and the validation
767 scope supported by the available field design.

768 The Lovelock field case reinforces this distinction under a separate basin

769 fill pumping/recovery design. Its smaller reduction in response variance likely
 770 reflects differences in site scale, preprocessing, environmental correction, ob-
 771 servation weighting, and the regional pumping/recovery design. The parallel
 772 diagnostic display interprets Lovelock using the same log model factor disper-
 773 sion, model factor distribution, BIC support, and parameter-variability met-
 774 rics used for Massachusetts. The evidence direction remains consistent: lag-
 775 ging Darcy reduces log model factor dispersion and remains competitive after
 776 model complexity penalties, while storage and transmissivity do not converge
 777 uniformly. This case shows that unresolved temporal response structures re-
 778 main visible across contrasting field settings and materially change inferred
 779 parameter values.

780 *3.11. Relation to stochastic hydrogeology and model uncertainty*

781 The proposed interpretation analysis complements stochastic hydrogeol-
 782 ogy and groundwater model uncertainty analysis rather than replacing them.
 783 Random field models and Monte Carlo simulations remain appropriate for
 784 propagating spatial heterogeneity through groundwater predictions (Dagan,
 785 1989; Gelhar, 1993; Rubin, 2003). Multimodel approaches remain appropri-
 786 ate when several plausible conceptual models require integration into predic-
 787 tions (Poeter and Anderson, 2005; Rojas et al., 2008; Ye et al., 2010). Broader
 788 groundwater uncertainty frameworks address model structure error, hydros-
 789 tratigraphic alternatives, conceptual model propositions, data worth, man-
 790 agement forecasts, and model structural adequacy (Refsgaard et al., 2006;

791 Li and Tsai, 2009; Tsai, 2010; Tsai and Elshall, 2013; Guillaume et al., 2016;
792 Fienen et al., 2025). The present work examines the preceding interpretation
793 step: the uncertainty introduced by drawdown-to-parameter interpretation
794 before investigators build a spatial field, model ensemble, or management
795 forecast.

796 Adjacent groundwater studies already address several downstream or
797 support-scale questions. Beckie and Harvey (Beckie and Harvey, 2002) show
798 that a slug test can be interpreted as a weighted support average. Manewell,
799 Doherty, and Hayes (Manewell et al., 2023, 2024) examine how aquifer test
800 interpretations can be incorporated into groundwater model parameteriza-
801 tion. Naderi and Gupta (Naderi and Gupta, 2023) define requirements for
802 inferring aquifer-scale transmissivity and storage in heterogeneous confined
803 aquifers. Li and Tsai (Li and Tsai, 2009), Tsai (Tsai, 2010), and Tsai and
804 Elshall (Tsai and Elshall, 2013) treat multimodel and hierarchical uncertainty
805 after groundwater model alternatives have been specified. The present pro-
806 cedure is placed before those downstream analyses. It evaluates whether the
807 parameters handed to them already carry unresolved interpretation depen-
808 dence.

809 The 10,000 controlled scenarios sharpen this distinction before the field
810 cases are used. Response residuals, parameter model factors, and decision
811 model factors do not have the same controlling drivers. Within the screened
812 records, response residual standard deviation is dominated by water-level
813 noise and observation support, whereas transmissivity and storage model fac-

814 tors remain strongly interpretation dependent. This behavior is important
 815 for BMA and groundwater management calculations. BMA can combine
 816 candidate predictions and separate uncertainty components, but the spread
 817 available to that calculation depends on the aquifer-test inputs supplied to
 818 it. The present diagnostic therefore propagates the four analytical interpre-
 819 tations into T , $D = T/S$, and $t_c \propto S/T$. Under variance window BMA, the
 820 selected interpretation can understate hydraulic control capacity by factors of
 821 1.60 and 1.18 at the median, and response time can be optimistic by factors of
 822 1.85 and 4.05. Later uncertainty propagation may therefore become overcon-
 823 fident if transmissivity and storage enter a numerical model as deterministic
 824 values, tight priors, or calibration labels without recording the interpretation
 825 model, log model factor, benchmark model factors, and field-to-benchmark
 826 applicability criteria. The proposed reporting structure improves traceabil-
 827 ity by linking measured drawdown, interpretation model, response-level log
 828 model factor, parameter spread, numerical model factor calibration, model
 829 averaged management diagnostic, and the decision quantity that inherits the
 830 parameter.

831 *3.12. Scope of inference from field data*

832 The two field cases define both the value and the scope of the study.
 833 The Massachusetts and Lovelock tests show that unresolved temporal re-
 834 sponse structure can materially change fitted responses and inferred parame-
 835 ters across contrasting geological settings. The numerical benchmark results

836 provide conditional external model factors for selected scenario classes, while
837 independent recovery of true field transmissivity, storage, leakage factor, or
838 lagging response times would require additional field constraints at the sup-
839 port scale. The analytical and numerical consistency warnings also define
840 practical limits: near well finite leakage behavior and coarse grid spacing
841 require additional checks before supporting strong field parameter claims.
842 Lagging Darcy is therefore treated as one candidate interpretation family,
843 while Theis, Hantush–Jacob leaky, and boundary diagnostics remain com-
844 peting conventional references.

845 The field response improvement establishes a diagnostic role for lagging
846 Darcy within broader aquifer-test interpretation. Project design use requires
847 the same evidence demanded of conventional interpretations: independent
848 constraints, blocked predictive validation, no major identifiability failure,
849 and decision propagation for the specific project. The field cases reach a
850 diagnostic standard and define the evidence needed before project specific
851 design use.

852 *3.13. Groundwater calculation inheritance*

853 The practical contribution is a reporting procedure for groundwater use
854 of aquifer test parameters. Estimates from pumping tests commonly enter
855 groundwater flow models, capture analyses, dewatering designs, contami-
856 nant control evaluations, and groundwater allocation studies as fixed input
857 parameters. The results indicate that reports should state the interpreta-

tion model, log model factor dispersion, support from information criteria, variability in inferred hydraulic properties, leakage or boundary alternatives when supported, and the supported scope of validation before treating fitted transmissivity and storage values as direct evidence of subsurface heterogeneity. This distinction supports defensible parameter selection and uncertainty communication when water resource decisions depend on groundwater response predictions.

The procedure is relevant when groundwater boundary conditions, pumping demands, or recharge conditions may change over a management or project horizon. Groundwater control and management problems often ask how much pumping capacity is needed and how quickly hydraulic response reaches monitoring points, supply wells, stream boundaries, or sensitive receptors. In those calculations, the relevant inherited quantities are not only drawdown residuals; they are T , S , T/S , and S/T . The model averaged diagnostic gives this effect a practical scale. Under variance window BMA, the 95th percentile robust capacity factor reaches 2.73 in Massachusetts and 4.52 in Lovelock Valley, and the 95th percentile response time factor reaches 3.32 and 8.79. The result defines an applicability check for later project specific optimization or management models: the pumping test report should document whether the selected interpretation leaves enough transformation uncertainty to alter the capacity or timing variables inherited by later groundwater calculations.

The field cases and numerical benchmark identify the set of diagnostics

needed before propagating aquifer test parameters to a project specific de-
watering, capture, saltwater intrusion, streamflow depletion, or groundwater
allocation model: log model factor dispersion, complexity support, param-
eter spread, leakage or boundary alternatives, scenario-conditioned model
factor COV, scope of validation, model averaged management diagnostic,
and the interpretation model used to estimate apparent aquifer properties
from measured drawdown.

3.14. Scope of the evidence

The evidence supports five conclusions. Pumping test interpretation can
introduce transformation uncertainty before transmissivity and storage are
propagated. The response-level log model factor quantifies interpretation-
dependent residual structure. Controlled numerical scenarios estimate con-
ditional parameter model factor bias and COV for specified aquifer classes.
Apparent variability in transmissivity and storage is partly dependent on
the interpretation model. The model averaged diagnostic shows that this
interpretation dependence can propagate to management-relevant hydraulic
quantities.

The present results support lagging Darcy as a diagnostic interpretation
family within a broader set of conventional aquifer-test solutions. They com-
plement existing groundwater uncertainty methods by examining the preced-
ing drawdown-to-parameter conversion. Field data alone provide response
diagnostics under specified observation supports; the numerical benchmark

903 supplies the scenario-conditioned controlled checks used to interpret param-
904 eter model factors.

905 3.15. *Limitations*

906 The field demonstration has a defined scope. The Massachusetts results
907 derive from one retained constant rate subset of a fractured bedrock pump-
908 ing test with 12 wells, and the Lovelock case provides an independent field
909 comparison with different preprocessing, observation weighting, and support.
910 The reported field COV values are specific to these test systems. The nu-
911 merical benchmark model factor COV values are conditional on the sce-
912 nario class, support definition, observation geometry, pumping schedule, grid
913 design, screening criteria, and four implemented analytical interpretations.
914 Their use as transformation priors is appropriate when field diagnostics sup-
915 port a comparable class. Direct evaluation of field parameter recovery would
916 require independent reference values for T , S , B , r_I , τ_q , and τ_s . Evaluation of
917 the full second moment decomposition in Equation 3 would also require com-
918 plete measurement budgets for water levels, pumping rates, distances, aquifer
919 thicknesses, and the simplification of the Massachusetts pumping history with
920 variable rates.

921 The model averaged management analysis uses fitted aquifer test param-
922 eters and analytical decision indices to define an inheritance diagnostic for
923 later groundwater calculations. Project specific groundwater management
924 models, capture zone analyses, or optimization studies would provide the

925 next level of decision support. The variance window weights are a sensitiv-
 926 ity diagnostic and should be read alongside the strict BIC weights rather
 927 than as a unique posterior probability claim. The final MAP refit uses weak
 928 lognormal prior residuals, so the lagging time and leakage estimates are reg-
 929 ularized diagnostic response quantities in the current field evidence. Several
 930 lagging leaky fits approach the upper bound for inactive leakage in B , in-
 931 dicating weak identifiability of the leakage factor in those wells. The local
 932 MAP covariance diagnostic relies on a quadratic approximation around the
 933 fitted optima. The Hantush–Jacob and constant head boundary fits repre-
 934 sent diagnostic equivalent interpretations of leakage or boundary behavior.
 935 The validation that excludes one well at a time operates as a diagnostic for
 936 spatial transfer in the Massachusetts case, while the Lovelock data provide
 937 the independent field consistency check.

938 The benchmark-transfer analysis is bounded by the numerical design.
 939 Its validation supports prediction of model-factor magnitude better than
 940 unique scenario identification, so the transferred COV values are reported as
 941 benchmark-conditioned sensitivity brackets for field diagnostics represented
 942 by the benchmark.

943 Future work should add independent reference tests and broader model
 944 families, including delayed yield, wellbore storage and skin, generalized ra-
 945 dial flow, dual porosity, exchange between fractures and matrix, unconfined
 946 numerical benchmarks, and heterogeneous numerical inverse models. Those
 947 additions would strengthen the current field evidence through explicit sup-

948 port criteria, blocked validation designs, profile likelihood or posterior checks,
949 and propagation to dewatering, capture, saltwater intrusion, or groundwater
950 allocation criteria for specific projects.

951 4. Conclusions

952 The analysis establishes a benchmark-conditioned framework for trac-
953 ing how aquifer-test drawdown becomes inferred groundwater parameters.
954 The framework separates measured response, four analytical transformations,
955 response-level log model factors, support-matched numerical reference values,
956 benchmark model-factor magnitude, field transfer diagnostics, and decision
957 quantities inherited by later groundwater calculations. This sequence shifts
958 the uncertainty problem upstream of conventional stochastic groundwater
959 modeling: transmissivity and storage enter later models as interpreted quan-
960 tities, not direct observations.

961 The 10,000 scenario numerical benchmark provides the controlled refer-
962 ence basis. It produced 160,012 fitted interpretation and well records across
963 homogeneous, leaky, finite-boundary, heterogeneous, and combined settings.
964 Analytical limiting checks confirmed physical consistency in 22 of 25 cases,
965 and numerical consistency checks aligned with analytical reference responses
966 in 30 of 32 configurations. The screened benchmark showed that response
967 fit, parameter recovery, and decision-factor propagation are distinct quanti-
968 ties; across class and interpretation groups in the screened records, median
969 model-factor COV values were 2.06 for transmissivity, 1.27 for storage, and

970 1.21 for response time.

971 The benchmark-transfer analysis links that controlled benchmark to field
972 diagnostics while preserving ambiguity among possible aquifer-process fam-
973 ilies. Nonlinear regression ensembles by complete numerical scenario pre-
974 dicted the 90th percentile absolute log model factor with cross-validated
975 R^2 values of 0.48, 0.51, and 0.55 for transmissivity, storage, and response
976 time. Applying the same response diagnostics to the field cases gave p90-
977 equivalent benchmark-transferred model-factor COV values of 0.61–0.90 for
978 Massachusetts and 0.31–0.51 for Lovelock Valley. These values are sensitivity
979 brackets conditioned on the represented benchmark diagnostics.

980 The two field cases demonstrate how the benchmark-conditioned frame-
981 work changes aquifer-test reporting. In Massachusetts, pooled log model
982 factor dispersion decreased from 0.166 for Theis confined to 0.062 for lagging
983 Darcy without leakage and 0.035 for lagging Darcy with leakage. In Lovelock
984 Valley, dispersion decreased from 0.246 for Theis confined to approximately
985 0.174 for the two lagging interpretations, while BIC support concentrated in
986 the no-leakage lagging form. These results support lagging Darcy as a useful
987 diagnostic interpretation for unresolved temporal response structure within
988 a broader set of conventional pumping-test solutions.

989 The model-averaged management diagnostic shows why the upstream
990 drawdown-to-parameter conversion matters for groundwater calculations.
991 Under variance window weighting of the four formal interpretations, the
992 median robust hydraulic control capacity factor was 1.60 in Massachusetts

993 and 1.18 in Lovelock Valley, with 95th percentile factors of 2.73 and 4.52.
 994 The median response-time factor was 1.85 and 4.05, with 95th percentile
 995 factors of 3.32 and 8.79. Aquifer-test reports should therefore state the
 996 interpretation model, response residual structure, benchmark transfer sup-
 997 port, parameter spread, validation scope, and inherited decision consequence
 998 before fitted parameters are treated as direct hydrogeologic evidence or
 999 deterministic groundwater model inputs.

1000 **CRedit authorship contribution statement**

1001 Ying-Fan Lin: Conceptualization, methodology, software, validation,
 1002 formal analysis, investigation, data curation, visualization, writing–original
 1003 draft, writing–review and editing, funding acquisition.

1004 **Acknowledgments**

1005 YFL thanks the John Su Foundation for its support.

1006 **Nomenclature**

Symbol	Meaning
S	Storage coefficient inferred from pumping test interpretation
T	Transmissivity inferred from pumping test interpretation
B	Leakage factor in the Hantush–Jacob or lagging leaky interpretation
r_I	Effective image well distance in the constant head boundary diagnostic
τ_q	Diagnostic flux gradient lag time in the lagging Darcy interpretation
τ_s	Diagnostic storage release lag time in the lagging Darcy interpretation
ϵ	Transformation uncertainty or residual component in the response model
c	Case specific positive offset used in the log response transformation

Symbol	Meaning
G	Diagnostic group of observations, such as full record, well, field case, or time window
$z_{\text{obs},i,j}, z_{M,i,j}$	Observed and modeled log responses for well i and time j
$\eta_{M,i,j}$	Log model factor for interpretation model M , well i , and observation time j
$\hat{F}_{\eta,M}$	Empirical distribution of the log model factor for interpretation model M
$F_{M,i,j}$	Multiplicative model factor, equal to $\exp(\eta_{M,i,j})$
$M_{T,M}, M_{S,M}$	Numerical benchmark parameter model factors for transmissivity and storage under interpretation model M
$T_{\text{app},M}, S_{\text{app},M}$	Apparent transmissivity and storage inferred by interpretation model M
$T_{\text{ref}}, S_{\text{ref}}$	Support-matched reference transmissivity and storage in the numerical benchmark
COV_M	Coefficient of variation of a lognormal model factor computed from $\sigma_{\ln M}$
$w_{M,i}$	BIC posterior weight for interpretation model M in field case or well group i
$w_{M,i}^V$	Variance window BMA weight for interpretation model M
$\Delta BIC_{M,i}$	Difference between model M and the minimum BIC value for well or group i
$s_{\Delta BIC,i}$	Standard deviation of the four ΔBIC values used in the variance window weight
$x_{M,i}$	Model-derived decision variable entering the model averaged management diagnostic
$D_{M,i}$	Hydraulic diffusivity, equal to $T_{M,i}/S_{M,i}$
$t_{c,M,i}$	Characteristic response time proportional to $S_{M,i}/T_{M,i}$ at distance r_i
M_i^*	Selected interpretation under the variance window BMA weight
R_i^C	Robust hydraulic control capacity factor relative to the selected interpretation
P_i^C	Probability that the selected interpretation underestimates hydraulic control capacity
R_i^t	Response time factor relative to the selected interpretation
P_i^t	Probability that the BMA mixture contains a faster response than the selected interpretation
SD_ϵ	Standard deviation of the log model factor in a diagnostic group
$COV_{F,M}$	Lognormal coefficient of variation for the multiplicative response model factor

Symbol	Meaning
VR_ϵ	Response variance ratio between lagging and classical transformations
ξ_d	Derived or design property produced by a transformation model
ξ_m	Measured property or measured response entering a transformation model
$\mathbf{y}_{m,i}$	Measured pumping test vector for observation well i
$\boldsymbol{\theta}_{M,i}$	Parameter vector inferred by interpretation model M for well i
$\mathcal{T}, \mathcal{T}_M$	Generic transformation function and interpretation transformation associated with model M
COV_X	Lognormal coefficient of variation for parameter X
VR_X	Apparent spatial variance ratio for parameter X
AIC_M, BIC_M	Akaike and Bayesian information criteria for model M

1007 Code availability

1008 Name of code/library: Aquifer-test transformation uncertainty bench-
1009 mark workflow.

1010 Developer and contact: Ying-Fan Lin, Department of Civil Engineering,
1011 Chung Yuan Christian University, yflin1110@cycu.edu.tw.

1012 Hardware requirements: Standard Windows, Linux, or macOS worksta-
1013 tion for the analytical and figure-generation workflow; the complete 10,000
1014 scenario numerical benchmark benefits from a multi-core CPU and at least
1015 16 GB RAM. GPU acceleration is not required.

1016 Program language: Python 3 and LaTeX. MODFLOW 6 is used as the
1017 numerical groundwater-flow simulator through Python workflow scripts.

1018 Software required: Python packages used by the scripts include NumPy,
1019 pandas, SciPy, scikit-learn, matplotlib, cmcrameri, FloPy, and Pillow. La-
1020 TeX compilation uses the Elsevier elsarticle class and BibTeX.

1021 Program size: The analysis scripts are approximately 0.47 MB. The pub-
1022 lic GitHub repository contains the manuscript source, figure-generation and
1023 analysis scripts, selected summary tables, and reproducibility instructions.
1024 The full generated benchmark and field-analysis tables are larger than 200
1025 MB and are archived in HydroShare.

1026 Source code: [https://github.com/aar246860/aquifer-test-tu-b](https://github.com/aar246860/aquifer-test-tu-benchmark)
1027 **enchmark**. The full reproducible data and software archive is available in
1028 HydroShare at [https://www.hydroshare.org/resource/3e8de7ff140c4f](https://www.hydroshare.org/resource/3e8de7ff140c4ffab03163e215bb6634/)
1029 **fab03163e215bb6634/**.

1030 **Data availability**

1031 The data set and reproducible analysis archive supporting the analyses
1032 are available in HydroShare (Lin, 2026). The resource includes analysis code,
1033 copied Massachusetts pumping-test inputs, processed Lovelock Valley tables,
1034 MODFLOW 6 numerical benchmark tables, analytical verification tables,
1035 field diagnostic result tables, figures, and summary tables needed to repro-
1036 duce the analyses. It is shared under the Creative Commons Attribution 4.0
1037 International license (CC BY 4.0; [https://creativecommons.org/licens](https://creativecommons.org/licenses/by/4.0/)
1038 **es/by/4.0/**). The Lovelock Valley raw data were originally published by the
1039 U.S. Geological Survey and are cited separately in this manuscript (Nadler,
1040 2020b).

1041 **Declarations**

1042 *Ethics approval*

1043 Not applicable. The analysis uses published field data and numerical
1044 benchmark simulations and does not involve human participants, human
1045 data, animals, or biological material.

1046 *Consent to participate*

1047 Not applicable.

1048 *Consent for publication*

1049 Not applicable.

1050 *Competing interests*

1051 The author has no relevant financial or non-financial interests to disclose.

1052 *Funding*

1053 This work was supported by the Taiwan National Science and Technology
1054 Council under grant numbers 113-2222-E-033-002-MY2 and 114-2221-E-033-
1055 005. The funders had no role in study design, data collection, analysis and
1056 interpretation, manuscript preparation, or the decision to submit the article
1057 for publication.

1058 *Declaration of AI-assisted language editing*

1059 AI-assisted language tools were used only for English editing, consistency
1060 checks, and manuscript format conversion. The author reviewed and edited
1061 the content and takes full responsibility for the submitted manuscript.

References

- Akaike, H., 1974. A new look at the statistical model identification. IEEE Transactions on Automatic Control 19, 716–723. doi:10.1109/TAC.1974.1100705.
- Anderson, M.P., Woessner, W.W., Hunt, R.J., 2015. Applied Groundwater Modeling: Simulation of Flow and Advective Transport. 2 ed., Academic Press.
- Bakker, M., Post, V., Langevin, C.D., Hughes, J.D., White, J.T., Starn, J.J., Fienen, M.N., 2016. Scripting MODFLOW model development using Python and FloPy. Groundwater 54, 733–739. doi:10.1111/gwat.12413.
- Barker, J.A., 1988. A generalized radial flow model for hydraulic tests in fractured rock. Water Resources Research 24, 1796–1804. doi:10.1029/WR024i010p01796.
- Bear, J., 1979. Hydraulics of Groundwater. McGraw-Hill, New York.
- Beckie, R., Harvey, C.F., 2002. What does a slug test measure: An investigation of instrument response and the effects of heterogeneity. Water Resources Research 38, 1290. doi:10.1029/2001WR001072.
- Beven, K., Binley, A., 1992. The future of distributed models: Model calibration and uncertainty prediction. Hydrological Processes 6, 279–298. doi:10.1002/hyp.3360060305.

- Boulton, N.S., 1954. The drawdown of the water-table under non-steady conditions near a pumped well in an unconfined formation. *Proceedings of the Institution of Civil Engineers* 3, 564–579. doi:10.1680/ipeds.1954.12586.
- Carrera, J., Neuman, S.P., 1986. Estimation of aquifer parameters under transient and steady state conditions: 1. maximum likelihood method incorporating prior information. *Water Resources Research* 22, 199–210. doi:10.1029/WR022i002p00199.
- Cho, H.J., Fiacco, Jr., R.J., Daly, M.H., 2004. Pumping test analysis in a fractured crystalline bedrock, in: *Proceedings of the 2004 US EPA/National Ground Water Association Fractured Rock Conference: State of the Science and Measuring Success in Remediation*, pp. 161–172.
- Cho, H.J., Fiacco, Jr., R.J., Daly, M.H., 2008. Characterization of crystalline bedrock contaminated by dense nonaqueous liquid. *Ground Water Monitoring & Remediation* 28, 49–59.
- Cooper, H.H., Jacob, C.E., 1946. A generalized graphical method for evaluating formation constants and summarizing well-field history. *Eos, Transactions American Geophysical Union* 27, 526–534. doi:10.1029/TR027i004p00526.
- Dagan, G., 1989. *Flow and Transport in Porous Formations*. Springer, Berlin. doi:10.1007/978-3-642-75015-1.

- Doherty, J., 2015. Calibration and Uncertainty Analysis for Complex Environmental Models. Watermark Numerical Computing.
- Fienen, M.N., Schachter, L.A., Hunt, R.J., 2025. A model uncertainty quantification protocol for evaluating the value of observation data. Scientific Investigations Report 2025-5007. U.S. Geological Survey. doi:10.3133/sir20255007.
- Freeze, R.A., Cherry, J.A., 1979. Groundwater. Prentice Hall, Englewood Cliffs, New Jersey.
- Gelhar, L.W., 1993. Stochastic Subsurface Hydrology. Prentice Hall, Englewood Cliffs, New Jersey.
- Guillaume, J.H.A., Hunt, R.J., Comunian, A., Fu, B., Blakers, R.S., 2016. Methods for exploring uncertainty in groundwater management predictions, in: Integrated Groundwater Management. Springer, pp. 711–737. doi:10.1007/978-3-319-23576-9_28.
- Gupta, H.V., Clark, M.P., Vrugt, J.A., Abramowitz, G., Ye, M., 2012. Towards a comprehensive assessment of model structural adequacy. Water Resources Research 48, W08301. doi:10.1029/2011WR011044.
- Hantush, M.S., Jacob, C.E., 1955. Non-steady radial flow in an infinite leaky aquifer. Eos, Transactions American Geophysical Union 36, 95–100. doi:10.1029/TR036i001p00095.

- Hill, M.C., Tiedeman, C.R., 2007. Effective Groundwater Model Calibration: With Analysis of Data, Sensitivities, Predictions, and Uncertainty. Wiley.
- Hoeting, J.A., Madigan, D., Raftery, A.E., Volinsky, C.T., 1999. Bayesian model averaging: A tutorial. *Statistical Science* 14, 382–417. doi:10.1214/ss/1009212519.
- Illman, W.A., Liu, X., Takeuchi, S., Yeh, T.C.J., Ando, K., Saegusa, H., 2009. Hydraulic tomography in fractured granite: Mizunami underground research site, Japan. *Water Resources Research* 45, W01406. doi:10.1029/2007WR006715.
- Kitanidis, P.K., 1995. Quasi-linear geostatistical theory for inversion. *Water Resources Research* 31, 2411–2419. doi:10.1029/95WR01945.
- Langevin, C.D., Hughes, J.D., Banta, E.R., Niswonger, R.G., Panday, S., Provost, A.M., 2017. Documentation for the MODFLOW 6 groundwater flow model. Techniques and Methods 6-A55. U.S. Geological Survey. doi:10.3133/tm6A55.
- Li, X., Tsai, F.T.C., 2009. Bayesian model averaging for groundwater head prediction and uncertainty analysis using multimodel and multimethod. *Water Resources Research* 45, W09403. doi:10.1029/2008WR007488.
- Lin, Y.C., Huang, C.S., Yeh, H.D., 2019. Analysis of unconfined flow induced by constant rate pumping based on the lagging theory. *Water Resources Research* 55, 3925–3940. doi:10.1029/2018WR023893.

- Lin, Y.C., Yeh, H.D., 2017. A lagging model for describing drawdown induced by a constant-rate pumping in a leaky confined aquifer. *Water Resources Research* 53, 8500–8511. doi:10.1002/2017WR021115.
- Lin, Y.F., 2026. Lagging Darcy diagnostic framework for pumping-test transformation uncertainty [dataset and software]. URL: <https://www.hydroshare.org/resource/3e8de7ff140c4ffab03163e215bb6634/>. creative Commons Attribution 4.0 International (CC BY 4.0).
- Liu, X., Illman, W.A., Craig, A.J., Zhu, J., Yeh, T.C.J., 2007. Laboratory sandbox validation of transient hydraulic tomography. *Water Resources Research* 43, W05404. doi:10.1029/2006WR005144.
- Manewell, N., Doherty, J., Hayes, P., 2024. Translating pumping test data into groundwater model parameters: A workflow to reveal aquifer heterogeneities and implications in regional model parameterization. *Frontiers in Water* 5, 1334022. doi:10.3389/frwa.2023.1334022.
- Manewell, N.M., Doherty, J., Hayes, P., 2023. Spatial averaging implied in aquifer test interpretation: The meaning of estimated hydraulic properties. *Frontiers in Earth Science* 10, 1079287. doi:10.3389/feart.2022.1079287.
- Naderi, M., Gupta, H.V., 2023. On the requirements for inferring aquifer-scale T and S in heterogeneous confined aquifers. *Water Resources Research* 59, e2023WR034430. doi:10.1029/2023WR034430.

- Nadler, C., 2020a. Analysis of aquifer framework and hydraulic properties of Lovelock Valley, Lovelock, Nevada. Open-File Report 2019-1133. U.S. Geological Survey. doi:10.3133/ofr20191133.
- Nadler, C.A., 2020b. Supplemental data for analysis of aquifer framework and hydraulic properties of Lovelock Valley, Lovelock, NV. doi:10.5066/P9LIL7PZ.
- Neuman, S.P., 1975. Analysis of pumping test data from anisotropic unconfined aquifers considering delayed gravity response. Water Resources Research 11, 329–342. doi:10.1029/WR011i002p00329.
- Papadopoulos, I.S., Cooper, H.H., 1967. Drawdown in a well of large diameter. Water Resources Research 3, 241–244. doi:10.1029/WR003i001p00241.
- Phoon, K.K., Kulhawy, F.H., 1999a. Characterization of geotechnical variability. Canadian Geotechnical Journal 36, 612–624. doi:10.1139/t99-038.
- Phoon, K.K., Kulhawy, F.H., 1999b. Evaluation of geotechnical property variability. Canadian Geotechnical Journal 36, 625–639. doi:10.1139/t99-039.
- Poeter, E., Anderson, D., 2005. Multimodel ranking and inference in ground water modeling. Groundwater 43, 597–605. doi:10.1111/j.1745-6584.2005.0061.x.

- Powers, J.P., Corwin, A.B., Schmall, P.C., Kaeck, W.E., 2007. Construction Dewatering and Groundwater Control: New Methods and Applications. 3 ed., John Wiley & Sons.
- Refsgaard, J.C., van der Sluijs, J.P., Brown, J., van der Keur, P., 2006. A framework for dealing with uncertainty due to model structure error. *Advances in Water Resources* 29, 1586–1597. doi:10.1016/j.advwatres.2005.11.013.
- Rojas, R., Feyen, L., Dassargues, A., 2008. Conceptual model uncertainty in groundwater modeling: Combining generalized likelihood uncertainty estimation and Bayesian model averaging. *Water Resources Research* 44, W12418. doi:10.1029/2008WR006908.
- Rubin, Y., 2003. *Applied Stochastic Hydrogeology*. Oxford University Press, New York.
- Sanchez-Vila, X., Meier, P.M., Carrera, J., 1999. Pumping tests in heterogeneous aquifers: An analytical study of what can be obtained from their interpretation using Jacob's method. *Water Resources Research* 35, 943–952. doi:10.1029/1999WR900007.
- Schwarz, G., 1978. Estimating the dimension of a model. *The Annals of Statistics* 6, 461–464. doi:10.1214/aos/1176344136.
- Stehfest, H., 1970. Algorithm 368: Numerical inversion of laplace transforms. *Communications of the ACM* 13, 47–49. doi:10.1145/361953.361969.

- Theis, C.V., 1935. The relation between the lowering of the piezometric surface and the rate and duration of discharge of a well using groundwater storage. Transactions, American Geophysical Union 16, 519–524. doi:10.1029/TR016i002p00519.
- Tsai, F.T.C., 2010. Bayesian model averaging assessment on groundwater management under model structure uncertainty. Stochastic Environmental Research and Risk Assessment 24, 845–861. doi:10.1007/s00477-010-0382-3.
- Tsai, F.T.C., Elshall, A.S., 2013. Hierarchical Bayesian model averaging for hydrostratigraphic modeling: Uncertainty segregation and comparative evaluation. Water Resources Research 49, 5520–5536. doi:10.1002/wrcr.20428.
- Vehtari, A., Gelman, A., Gabry, J., 2017. Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. Statistics and Computing 27, 1413–1432. doi:10.1007/s11222-016-9696-4.
- Warren, J.E., Root, P.J., 1963. The behavior of naturally fractured reservoirs. Society of Petroleum Engineers Journal 3, 245–255. doi:10.2118/426-PA.
- Xu, T., Valocchi, A.J., 2015. A Bayesian approach to improved calibration and prediction of groundwater models with structural error. Water Resources Research 51, 9290–9311. doi:10.1002/2015WR017912.

- Ye, M., Pohlmann, K.F., Chapman, J.B., Pohll, G.M., Reeves, D.M., 2010. A model-averaging method for assessing groundwater conceptual model uncertainty. *Groundwater* 48, 716–728. doi:10.1111/j.1745-6584.2009.00633.x.
- Yeh, T.C.J., Liu, S., 2000. Hydraulic tomography: Development of a new aquifer test method. *Water Resources Research* 36, 2095–2105. doi:10.1029/2000WR900114.
- Yeh, W.W.G., 1986. Review of parameter identification procedures in groundwater hydrology: The inverse problem. *Water Resources Research* 22, 95–108. doi:10.1029/WR022i002p00095.